A Unified Framework for Medicinal Leaf Classification Using Deep Learning and Recommendation Using TF-IDF and Logistic Regression

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CERTIFICATE

This is to certify that this project report titled "**A Unified Framework for Medicinal Leaf Classification Using Deep Learning and Recommendation Using TF-IDF and Logistic Regression**" is a Bonafide record of work done by **K.TENIA (228W1A12G6)** and **A.MALLIKHARJUNA RAO (228W1A12D3)** under my guidance and supervision, submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V. R. Siddhartha Engineering College** (Autonomous under JNTUK) during the academic year 2025-26.

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PROJECT SUMMARY

S.No	Item	Description
1	**Project Title**	**A Unified Framework for Medicinal Leaf Classification Using Deep Learning and Recommendation Using TF-IDF and Logistic Regression**
2	**Student Names & Numbers**	**K. TENIA (228W1A12G6)** **A. MALLIKHARJUNA RAO (228W1A12D3)**
3	**Name of The Guide**	**DR.T. ANURADHA, Professor**
4	**Research Group**	**Deep Learning (DL), Machine Learning (ML), Natural Language Processing (NLP), Artificial Intelligence (AI)**
5	**Application Area**	**Computational Ethnobotany & AI-Assisted Healthcare**
6	**Aim of the Project**	**To create an intelligent system capable of recognizing the leaves of medicinal plants and offering information regarding their medicinal properties. Deep learning technology will be used to recognize images and NLP techniques to suggest medicinal usage for symptoms described by the users. The system will be able to give immediate information about the usage and dosage and also about precautions.**
7	**Project Outcomes**	**1. A deep learning classifier (ResNet-50) was designed for effective and reliable classification of medicinal leaves.** **2. Designed an NLP based system (TF-IDF + Logistic Regression) to identify best medicinal leaves with usage, dosage and precautions based on symptoms.** **3. Created a full-fledged mobile application with user interactions through images and text.**

Student Signatures

1.

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Signature of the Guide

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ABSTRACT

The fast extinction of traditional botany, in addition to the growing demand worldwide for herbal medicine, creates an urgent need for efficient and intelligent systems for identifying and recommending medicinal plants. Manual identification is not always reliable because of morphological resemblance among various plants. The medicinal knowledge is fragmented and stored in various informal places, making it difficult to be accessed by everyone.

This project is proposing a comprehensive AI-based framework combining Deep Learning and Natural Language Processing to solve the problem. At first, an image classifier model based on ResNet-50 architecture using transfer learning was created, which can detect 80 Indian medicinal plant species using their leaves' pictures. The model runs in the FastAPI backend hosted on Hugging Face Spaces and embedded in a React Native mobile app that gives real-time predictions with medicinal information.

Finally, a recommender module based on NLP technology has been added. A custom dataset was used consisting of fields such as Diseases, Symptoms, Leaf name, Uses, Dosages, and Precautions. A TF-IDF vectorizer and logistic regression model predicts the top three suitable medicinal plants depending on user input symptoms or diseases.

This approach allows text and image input using a stateless interaction system like a chatbot. The image classification model is about 97 percent accurate while the NLP engine provides relevant recommendations, thereby forming an interface that integrates traditional medicine with modern AI solutions.

Keywords:

Medicinal Plant Identification, Deep Learning, ResNet-50, Transfer Learning, Natural Language Processing, TF-IDF, Logistic Regression, Text Classification, Leaf Recommendation System, FastAPI, React Native, Mobile Application, AI in Healthcare

CHAPTER – 1
INTRODUCTION

1.1 Origin of the Problem

India is known to be "the Botanical Garden of the World," since there is a high level of diversity in this country where different kinds of medicinal plants grow in varying climatic zones [11]. India flora served as a basis for the creation of Ayurvedic, Siddha, and Unani traditional healing systems throughout many years. These medical systems used various plants together with some plant parts, for example, leaves, roots, barks, or seeds, to treat some disease [10]. It was possible to know about medicinal properties of plants because of generations' transfer of information from ancestors to younger people by oral communication or practice of herbal healing. Knowledge about plant morphology and characteristic features like leaves shape, venation, smell, and others made it possible to understand how one can heal using plants [15]. Plants could serve to cure simple colds, skin diseases, digestion problems, and serious illnesses.

However, in this era, there is a growing concern that poses a threat to the sustainability of this vast knowledge. Urbanization and rural-to-urban migration rates have become high, and this has resulted in the interruption of the transmission process of ethnobotanical knowledge in the communities. While learning institutions emphasize science and technology education, the significance of indigenous knowledge system that leads to separation from nature has been overlooked. Young people nowadays find it hard to connect with plant species found in their surroundings and how these species are effective in curing diseases. This situation is referred to as "plant blindness," whereby people can easily identify manufactured products and artificial designs rather than native plants in their surroundings [11]. Despite the growing interest in herbal and organic products worldwide due to the increased knowledge about natural medicine, there is one crucial ability that is still missing, and that is the identification of medicinal plants.

The paradox poses a significant problem. Even though there has been increasing interest in herbal remedies, the fundamental facts about using them are lacking. This could prove damaging because the incorrect identification of herbs may not only render them ineffective

but might even pose risks to human health. In addition, the wealth of information available within herbal medicines is dispersed in ancient books, scientific literature, and folklore, making it difficult to utilize it effectively [11].

However, the development of this system starts from solving this problem with the help of technologies from the field of Artificial Intelligence to bring traditional knowledge about plants and medicine into digital format. One of the first steps of this project is associated with the imitation of professional skills of botanists in identifying different medical leaves with the help of deep learning algorithms [1], [9]. However, having in mind that questions related to health care usually come from patients experiencing symptoms and not images, the creation of intelligent recommendation systems is another step that should be taken. Eventually, at the final stage, this system will be able to recognize and give recommendations based on symptoms [17], [20].

By integrating computer vision and natural language processing into a single platform, this initiative manages to ensure that the combined wisdom of the taxonomist and the herbal medicine expert is well-transmitted to every individual user using the mobile app [3], [4], [5]. While users may decide to identify plant species either by physical appearance or symptoms, they get access to much more than simply knowing the plant's name. They receive detailed instructions on how to consume the medicinal leaf and at which dose. By moving from data gathering to building an intelligent system, it becomes apparent that a critical stride has been made toward preserving knowledge.

1.2 Basic Definition and Background

In order to conduct a proper analysis of the proposed solution presented within this paper, it is crucial to offer a detailed elaboration on the technical foundation as well as the computational methods, which underlie the entire concept. In the case of this research, Deep Learning (DL) – being a new generation approach of machine learning – ought to be viewed as the main pillar. Specifically, Deep Learning is a subfield of machine learning, implying the application of artificial neural networks that have multiple hidden layers for detecting and forming complex and nonlinear dependencies between variables[1][2]. Here, the use of the word "deep" highlights the number of hidden layers involved in the process; thus,

allowing the algorithm to generate hierarchical representations of the input data. The distinguishing element of this methodology is its self-learning ability in relation to the selection of features, setting apart deep learning from conventional machine learning approaches, where the feature selection process is delegated to domain specialists [7].

Computer Vision forms the primary area of application of this study project. Computer Vision is a subfield in the broader area of Artificial Intelligence. Within this domain, scientists are aiming at developing computer systems that can interpret visual data through a multi-level analysis process. Visual data is represented digitally as a collection of numbers in the form of pixels. These numbers are analysed through CNNs, which help in recognizing visual patterns like edges, textures, and shapes [9]. The output from the CNNs after passing through multiple convolution and activation functions helps in generating a high-level abstraction for image recognition [10]. In this study project, the concept of computer vision has been applied to categorize plants based on the characteristics of their leaves [16].

One of the methodologies that enable this task is known as Transfer Learning [7] . Training deep convolutional neural networks generally requires large amounts of data, especially labeled data and computational resources; hence, it becomes difficult to train deep neural networks when data is scarce. The problem of having too little data for training deep neural networks can be solved using transfer learning in which one uses pre-trained models [7]. Pre-trained models are models that have previously been trained on large datasets like the ImageNet dataset that consists of millions of images belonging to different categories [2]. Pre-trained models learn about features like edge detection, texture recognition, shapes, etc. since these are domain-independent features, this makes the learning process easy. Here, we use ResNet-50, which is renowned for its residual learning technique and helps solve the vanishing gradient problem in training deep networks [1].

The technique used in this case is known as transfer learning. As such, the initial layers of ResNet-50 neural networks are not altered to ensure that their feature-extraction capacity from generalizable data is preserved, but the last layers are fine-tuned through learning using the particular data on medicinal leaves. This ensures efficiency in terms of time and cost without compromising on accuracy.

While computer vision is adopted as the foundation technology for the first model, the range of technologies being applied widens in the third stage to include Natural Language

Processing (NLP), which is adopted to handle the text-based input data in the final step of the project. In practice, it was found out that people would make interactions using their symptoms rather than images. This calls for a technology to comprehend the textual description of the disease or symptom. This would require the conversion of text to numerical data based on the idea of TF-IDF[18], [19].

Deep Learning technology applied to recognize images and classical machine learning methods used for textual classification contribute to the emergence of an artificial intelligence system based on two approaches. Not only does it help to recognize plants from images, but it also provides appropriate recommendations for medicinal leaves depending on user needs. This is proof that by applying different approaches in combination, one can successfully apply artificial intelligence to various areas of our life and create a very effective system for identifying and recommending medicinal plants.

1.3 Problem Statement

The main concern that this research aims to address is the considerable number of errors made while manually classifying the plants by untrained individuals, as well as the lack of credible data on how to use them for therapeutic purposes. In botany, one of the toughest challenges is morphological mimicry. Leaves from totally different plant families may share similar attributes like shape, vein pattern, coloration, and edges. Hence, differentiating them becomes difficult for a person who does not specialize in botanical studies. Even the leaves of plants that are known for their therapeutic value, like Justicia adhatoda, with its healing agents, can be mistaken for ordinary lanceolate leaves, which are not from medicinal plants but even poisonous ones. So, the impacts of the problem stated above should never be overlooked, considering the potential health risks involved. It is crucial to recognize that the misuse of species might lead to grave problems like allergies, poisoning, among others.

Besides these identification issues, one other important issue is that of disorganization and unattainability of medicinal information. Even after the correct plant being identified, the task of obtaining credible medicinal information regarding its uses, dosages, precautions, and everything else is not an easy task. Such information may be obtained from various sources, whether ancient Ayurvedic texts, scientific research, traditional knowledge, or even from less reliable online sources. Due to lack of a credible database where all such information has been consolidated, there are problems faced by those who seek it. Hence,

there is a clear gap between identification of plants and medicinal information.

Additionally, conventional methods rely mainly on visual identification and ignore another important scenario that prevails in reality: individuals might be attracted to traditional medicine depending on their symptoms or illnesses without having knowledge about the herbs. If someone feels like having fever, colds, or skin irritations, he/she cannot know what herb can treat him/her and even its look. Therefore, the present approaches do not take into account this user perspective.

These issues can be addressed by a combination approach where a system is developed that has the abilities of visual recognition as well as recommendation system. In the first phase of development, the system will need to be trained so as to work as an expert botanist who can identify various types of plants from their leaves. However, in order to overcome the issue of using only images to communicate with the system, in the final phase of development, a model based on NLP will be constructed so as to enable users to describe their symptoms and receive recommendations.

With such integration, the entire process will no longer be confined to being merely an identification tool for plants but rather will become an authorized pharmacology database. The integration will create a smart work process flow that will bridge the gap between botany and medicine in this modern age. The information from the database will be accurate and reliable, thereby providing safe and reliable use of the database.

1.4 Objectives & Outcomes

1. Objectives

The primary goals of the proposed project may be directly linked to the creation of an intelligent, scalable, and easy-to-use system for identification and recommendation of medicinal plants and their subsequent deployment through a mobile application.

- Development of a deep learning model using ResNet-50, which will be capable of performing classification of medicinal plants' leaves and providing medicinal information about the classified leaf.
- Development of a recommendation system using NLP techniques such as TF-IDF and logistic regression, which will be capable of analysing symptoms/diseases mentioned by

the user and recommending relevant medicinal leaves.

- Deploy both image classification and text recommendation applications within the backend architecture and test it via the cloud computing environment (Hugging Face Spaces).
- Create a mobile application using the React Native application development tool with image uploading capabilities along with the capability for stateless questions and answers.

2. Outcomes

The following results were achieved upon the completion of the project:

- Creation of a reliable deep learning classification model, utilizing ResNet-50, that is capable of classifying medicinal plants' leaves and identifying their therapeutic applications through databases effectively.
- Design and implementation of a recommendation model utilizing NLP, where a combination of the TF-IDF and Logistic Regression algorithms will be used to recommend medicinal plant leaves.
- Successful integration of visual classification and text-based recommendation models into a single backend model capable of operating in two modes.
- Use of the complete backend model in Hugging Face Spaces.
- Successful development of a mobile app for user convenience, facilitating both visual recognition and stateless Q&A.

1.5 Scope and Applications of the Project

The system's applicability is not confined to a single field alone but extends to various sectors such as society, education, health care, and industry. This renders the system highly adaptable and helpful in addressing the problem of connecting traditional knowledge with current technology. The integration of identification of plants from their photographs along with medicinal prescriptions from their symptoms helps achieve this objective.

- Educational Sector: The app acts as an innovative and interactive learning tool for learners/researchers studying areas like botany, Ayurveda, agriculture, forestry, and environmental science. The app acts as a digital field guide that will not only help to

identify plants using picture inputting but also helps learners understand the concept better by giving information on the medicinal benefits and use of the plant species in real time. Moreover, the addition of the NLP recommendation feature also facilitates learners to learn about the correlation between diseases, symptoms, and relevant medicines.

- The General Public and Home Remedies: This model will allow individuals to make sound judgments when using herbal medicines even though they have never had any experience in botany before. They can either decide to identify plants around them through the pictures or simply input their symptoms to receive adequate suggestions of which herbs they need to take. All these options will allow individuals to adopt a “back to nature” attitude by using nature to cure diseases, whether they be common colds, fevers, skin infections, or digestive problems.
- Healthcare and Preventive Medicine: The introduction of the symptom-based system enhances the use of the system in preventive healthcare. It is the initial decision-making system used by those interested in trying out other methods of treatment aside from traditional medicine. Without mentioning that this system cannot replace the opinion of experts, it offers useful help through structured information whereby people get to know which herbal remedy they need to take for their symptoms and the way they should take it..
- Pharmaceutical and Herbal Products: With respect to the sector concerned with the production of medicines by using pharmaceutical products and herbs, this model offers some advantages especially regarding quality and validation. The field workers who collect raw material for producing medicines will be able to validate plant type through the image recognition technique thereby minimizing the chances of adulteration. The researchers working on developing new medications from plant materials can benefit greatly from the symptom recognition system.
- Eco-Tourism and Forest Department: The application can be effectively used in botanical gardens, wildlife reserves, and national parks for creating an interactive platform among tourists and educating them about the biodiversity found in the area. All the tourists need to do is scan the plants and ask any questions related to the medicinal value of these plants. They will thus be able to gain knowledge about the

environment around them. This is an application that works as a guide for tourists to know information related to different types of plants.

In general terms, the proposed system has very good chances of having an impact on many domains because of the availability of an efficient and smart system for understanding and handling medicinal plant data.

CHAPTER - 2
REVIEW OF LITERATURE

2.1 Description of Existing Systems

Developing ideas about the project entailed going through existing literature related to computer vision and phenotyping. Phenotyping in plants has shifted from hand-crafted feature sets based on measurements of plant length and width to an end-to-end deep learning technique. These works have had significant impacts on our study, as discussed in table 2.1:

1. **"Deep Residual Learning for Image Recognition" (He et al., 2020) [1]**

   The work is a major breakthrough in the field of deep learning because it presented ResNet to address the degradation problem faced by very deep neural networks. The authors used residual blocks along with identity shortcuts, and were thus able to build neural networks with up to 152 layers, obtaining unprecedented levels of accuracy in key recognition tests. Their findings demonstrate the superiority of residual nets as they are relatively easy to optimize and attain improved accuracy through depth. Nonetheless, it can be seen that a limitation associated with ResNets, which was pointed out by other researchers, is their high depth, causing an elevated level of memory and computational overhead.

2. **"ImageNet Classification with Deep Convolutional Neural Networks" (Krizhevsky et al., 2022) [2]**

   This research paper serves as the bedrock for computer vision research today. This is because this paper introduced the concept of using AlexNet architecture to show that Deep Convolutional Neural Networks (DCNN) are capable of performing significantly better than any previous machine learning algorithm. This paper demonstrated that the DCNN was capable of learning features in a hierarchical manner from simple features like edges at the first level to more complex features like object parts in the subsequent levels. It is this capability to learn in a hierarchical manner that is crucial for leaf classification.

3. **"Medicinal Plant Recognition based on Vision Transformer and BEiT" (Nguyen Nhut et al., 2024)[3]**

   In the course of this research, the scientists employed the direct comparison of classic CNN models and new Transformer-based algorithms for performing medicinal plants identification. Using the VNPlant-200 data set, they compared classic models like EfficientNet with

Transformer models such as Bidirectional Encoder Image Transformer (BEiT) and other. The scientists found out that the most accurate results were provided by BEiT with the rate of 99.14%. At the same time, the issue of robustness was discovered, since the Transformer models proved to be less resistant to differences in lighting and background than CNN. Moreover, inferences took much more time in Transformer models compared with CNN. This data was particularly valuable for our project, confirming the choice of ResNet model

4. **"Multi-Label Plant Species Classification with Self-Supervised Vision Transformers" (Gustineli et al.,2024) [4]**

   In particular, the current study is regarded as the key milestone in using Vision Transformers in self-supervised tasks since the study applied the DINOv2 model structure. Researchers analysed various examples of plant recognition where multiple plants occurred in one picture. In view of the great computational load of these tasks, distributed data processing techniques like Apache Spark were applied. Nonetheless, the results demonstrated outstanding accuracy performance of the model; the literature review highlighted an interesting challenge related to the usage of Vision Transformers, namely these algorithms require high-resolution images for accurate analysis. Accordingly, Vision Transformers involve significant amounts of memory resources and long computing time, which means that Vision Transformers can be recommended only for scientific application in servers but not for quick mobile usage.

5. **"A Novel Approach to Navigate Taxonomic Hierarchy to Address Open-World Scenarios" (Sinha et al.,2025) [5]**

   The researchers Sinha, Rana, and Roy have proposed an extremely complex approach for the classification of plants not only into a particular species but also within the hierarchy of taxonomy (Family > Genus > Species) using the DenseNet121 architecture along with the Multi-Scale Self-Attention model. While being academically intriguing, the authors found that in this particular scenario, the “Open World” approach resulted in a trade-off between the accuracy of classification at a particular species level and generalization. Yet, since the primary prerequisite for any such system is the correct classification of the specific species, we found the current paper’s findings incompatible with hierarchical approaches.

6. "**Adam: A Method for Stochastic Optimization**" **(Kingma and Ba, 2023) [6]**

   This study stands as a major milestone in optimization as it introduced the Adam algorithm, which utilizes adaptive learning rates for training deep networks. The researchers combined

the benefits of AdaGrad and RMSProp to handle complex datasets effectively while maintaining low memory requirements. While the results showed exceptionally fast convergence across various tasks, the literature highlights a challenge regarding its generalization performance, which can sometimes be lower than standard Stochastic Gradient Descent. Consequently, Adam is highly recommended for efficient large-scale training on servers, though it may require extra tuning in scenarios where reaching the absolute peak accuracy is critical.

7. **"A Survey on Transfer Learning" (Pan and Yang, 2024) [7]**

   This influential survey is regarded as a foundational milestone as it provided the first systematic framework and taxonomy for transfer learning. The researchers explored how knowledge extracted from a source domain can be applied to a target task, significantly reducing the need for massive datasets. While the study demonstrates how pre-trained models boost learning efficiency, it highlights the challenge of "negative transfer," where using unrelated source data can actually decrease model performance. Consequently, transfer learning is highly recommended for scientific tasks with scarce data, provided the source and target domains are well-aligned. Thus, it remains a vital strategy for research-heavy applications where data acquisition is difficult or expensive.

8. "**Rethinking the Inception Architecture for Computer Vision**" **(Szegedy et al., 2022) [8]**

   This paper is a major milestone in computer vision because it introduced the Inception-v3 model, focusing on making deep networks much more efficient. The researchers found ways to break down large filters into smaller, factorized ones, allowing the model to be deeper without needing massive amounts of computing power. Their results set new records for accuracy, though the study points out that the architecture itself is quite complex and can be difficult to customize. Because of this complexity, Inception-v3 is great for heavy-duty server work, but it might be too demanding for simple mobile apps. It remains a go-to choice for scientific research where high-level accuracy is the top priority.

9. "**Deep-plant: Plant identification with convolutional neural networks**" **(Lee et al., 2021) [9]**

   This paper is a significant milestone as it pioneered the use of convolutional neural networks for automated plant identification, moving past the limitations of manual feature selection. By training deep models on leaf datasets, the researchers proved that AI could recognize species

with much higher precision than older methods. However, the study points out that these models are data-hungry and can struggle when leaf patterns across different species look nearly identical. Furthermore, the high computational cost means these algorithms are better for powerful desktop applications rather than quick use on standard mobile devices. Ultimately, this work is highly recommended for scientific botanical studies where high-level classification accuracy is essential.

10. **"Using Deep Learning for Image-Based Plant Disease Detection" (Mohanty et al., 2022) [10]**

    This study is seen as a major landmark in digital agriculture as it proved that deep learning can accurately identify dozens of plant diseases across various crop types. By testing models on the large PlantVillage dataset, the researchers achieved nearly 99% accuracy, showing just how powerful CNNs can be for farming. However, the study also highlights a real-world hurdle: the models worked perfectly in controlled lab settings but lost some precision when used in actual farm fields. Because these algorithms need high-quality data and strong processing power, they are currently better suited for scientific server applications than for quick, offline mobile apps. Ultimately, this work is a vital step toward automated crop monitoring, even if it requires more robust data for everyday use.

11. **"Plant Species Identification Using Computer Vision Techniques: A Systematic Literature Review" (Waldchen and Mader, 2023) [11]**

    This paper is a crucial milestone because it provides a complete look at how plant identification has evolved from manual feature extraction to modern deep learning. The researchers analyzed years of studies to show that while deep learning is the way forward, most models still struggle with the messy conditions of nature, like poor lighting or busy backgrounds. Even though accuracy is high in lab settings, the literature points out that real-world deployment remains a major challenge. As a result, these techniques are highly recommended for botanical research, but they need more field-testing before they can be fully trusted for casual mobile users. It remains a foundational guide for understanding the gap between current technology and practical, everyday application.

12. **"Xception: Deep Learning with Depthwise Separable Convolutions" (Chollet, 2024) [12]**

    This paper is a significant milestone because it introduced the Xception architecture, which rethinks how convolutions work by separating spatial and channel-wise processing. The

researchers found that this "extreme" version of the Inception model actually performs better and is more efficient with its parameters. Even though the results showed a clear leap in accuracy, the study suggests that the architecture still needs quite a bit of memory and processing speed to run well. As a result, Xception is a fantastic choice for heavy-duty image recognition on servers, but it might be a bit too much for standard mobile devices to handle. It stands as a vital development for anyone looking to push the limits of model performance.

13. "**MobileNetV2: Inverted Residuals and Linear Bottlenecks**" **(Sandler et al., 2025) [13]**

    This study is widely seen as a major milestone because it designed a neural network specifically for mobile devices without sacrificing too much accuracy. By introducing inverted residual blocks and linear bottlenecks, the researchers managed to make the model much lighter and faster than its predecessors. The results were impressive, showing that it's possible to get high-level performance even on hardware with limited power. However, the literature notes that while it is optimized for speed, it can still struggle with extremely fine details compared to massive, server-based architectures. Consequently, MobileNetV2 is highly recommended for real-time mobile applications and embedded systems where efficiency is just as important as accuracy.

14. "**PyTorch: An Imperative Style, High-Performance Deep Learning Library**" **(Paszke et al., 2023) [14]**

    This landmark paper is considered a major milestone because it proved that a deep learning library can be both easy to use and incredibly fast. The researchers introduced an "imperative" style where models are treated as regular Python code, allowing for much easier debugging and flexible changes on the fly. While the results showed it could match the speed of older, more rigid systems, the literature points out that its performance in massive, multi-node production environments still faced some initial scaling challenges. Consequently, PyTorch has become the top recommendation for research and prototyping on high-end servers, though it may require extra optimization steps for specialized mobile or embedded use. It remains a foundational tool that transformed how the scientific community develops new AI models.

15. "**Plant Leaf Classification using Shape and Texture Features**" **(Tiay et al., 2021) [15]**

    This study is regarded as a notable milestone in botanical imaging as it demonstrated how combining shape and texture features can significantly improve classification accuracy. The researchers focused on extracting detailed geometric traits and surface patterns to distinguish

between different leaf species more effectively. Although their results proved that this multi-feature approach is more reliable than using a single metric, the literature highlights the challenge of high computational complexity involved in manual feature extraction. Consequently, while these techniques are highly recommended for detailed scientific analysis on desktop systems, they are often too slow for real-time mobile identification. This work remains a key reference for understanding the value of hybrid feature sets in image recognition.

16. "**Deep Learning for Plant Identification in Wild Environments**" **(Sun et al., 2024) [16]**

    This paper is seen as a key milestone for taking plant identification out of the lab and into "wild" environments where backgrounds are messy and lighting varies. The researchers evaluated various deep learning models to see how they handle the visual noise of nature compared to controlled settings. Their findings showed that while deep models are highly effective at recognizing species in complex scenes, they require a massive amount of diverse training data to be truly reliable. However, the study notes that the high computational demand of these models makes it difficult to run them on standard mobile hardware without a strong internet connection. Consequently, this work is highly recommended for building server-side research tools, but it highlights the ongoing need for more efficient algorithms for field-based use by non-experts.

17. **"Comparative Analysis of Text Classification Algorithms" (Borges et al., 2025) [17]**

    This research presented a comprehensive comparison of various text classification algorithms, including Naïve Bayes, Support Vector Machines (SVM), and Decision Trees. The authors evaluated these models on standard textual datasets using performance metrics such as accuracy, precision, recall, and F1-score. Their findings indicated that while advanced models like SVM and ensemble methods achieved higher accuracy, simpler algorithms such as Naïve Bayes offered faster computation and were effective for smaller datasets. The study emphasized the trade-off between computational efficiency and classification performance, highlighting the importance of selecting models based on application requirements.

18. **"A Comparative Study on TF-IDF Feature Weighting Method and its Analysis using Unstructured Dataset" (Kowsari et al., 2024) [18]**

    This study focused on analysing the effectiveness of the TF-IDF feature weighting method for text classification tasks using unstructured datasets. The authors compared TF-IDF with other

feature extraction techniques to evaluate its impact on model performance. Their experiments demonstrated that TF-IDF is highly effective in handling sparse textual data and improves classification accuracy when used with traditional machine learning models. However, the study also identified a limitation: TF-IDF lacks the ability to capture semantic relationships between words, making it less effective compared to modern embedding techniques in complex language understanding tasks.

19. **"Comparative Analysis of TF-IDF and Word2Vec in Sentiment Analysis: A Case of Food Reviews" (Zhan et al., 2025) [19]**

    This research explored the performance differences between TF-IDF and Word2Vec representations in sentiment analysis of food review datasets. The authors trained machine learning classifiers using both feature extraction methods to evaluate their effectiveness. The results showed that Word2Vec outperformed TF-IDF in capturing semantic and contextual relationships, leading to improved classification accuracy. However, TF-IDF was found to be computationally simpler and more efficient, making it suitable for applications with limited resources. The study highlighted the trade-off between semantic understanding and computational efficiency

20. **"Robust Sentiment and Semantic Analysis of Small and Medium-Sized News Headline Datasets" (Liang et al., 2026) [20]**

    This study investigated sentiment and semantic analysis techniques on small and medium-sized datasets across domains such as sports, science, and agriculture. The authors employed a hybrid approach combining traditional natural language processing techniques with deep learning models to improve performance. Their findings revealed that hybrid models significantly enhanced accuracy and robustness, particularly when dealing with limited data. Additionally, domain-specific tuning was shown to further improve model effectiveness, emphasizing the importance of adapting models to specific contexts for better results.

2.2 Summary of literature

Table 2.1: Summary of Key Literature in Automated Plant Identification and symptom based ayurvedic solution

Authors (Year)	**Title of Paper /work**	**Description/Methodology**	**Key Contributions / findings relevant to this project**
He, K., Zhang, X., Ren, S., & Sun, J. (2020) [1]	Deep residual learning for image recognition	Suggested the ResNet network architecture that incorporates residual blocks, enabling training of very deep neural networks without encountering the problem of vanishing gradients.	Presented the ResNet50 model that is crucial for this research due to its effectiveness and relevance to transfer learning in image classification tasks.
A. Krizhevsky, I. Sutskever, G.E. Hinton (2022) [2]	ImageNet classification with deep CNNs	ImageNet Dataset has been introduced, which is a large set of images that help with pre-training of neural networks.	The ImageNet dataset is the source from which the pre-trained weights are derived for the ResNet50 model utilized in this research.
D. T. Nguyen Nhut, T. D. Tan, T. N. Quoc, and V. T. Hoang (2024) [3]	Medicinal plant recognition based on Vision Transformer and BEiT	The study implements a recognition framework for medicinal plants by comparing the performance of Vision Transformer (ViT) and Bidirectional Encoder representation from Image Transformers (BEiT). The methodology focuses on using self-supervised pre-training and transformer-based attention mechanisms to capture intricate leaf patterns and textures.	The research establishes that Transformer architectures, particularly BEiT, achieve superior accuracy over traditional CNNs for medicinal species. It highlights the effectiveness of self-supervised learning in handling complex botanical features, providing a highly reliable model for specialized pharmaceutical and research applications.
M. Gustineli, A. Miyaguchi, and I. Stalter (2024) [4]	Multi-Label Plant Species Classification with Self-	This study applies the DINOv2 self-supervised model structure to handle multi-label plant	The paper marks a key milestone in using self-supervised Vision Transformers for

	Supervised Vision Transformers	recognition, where a single image may contain multiple different species. To manage the massive computational load required for high-resolution image analysis, the researchers implemented distributed data processing techniques using Apache Spark. .	multi-label tasks, achieving outstanding accuracy. However, it also highlights that the high memory and computing time required for these high-resolution models make them better suited for server-side scientific applications rather than mobile use.
S. Sinha, T. Rana, and R. Roy (2025) [5]	A novel approach to navigate the taxonomic hierarchy to address Open-World Scenarios in Medicinal Plant Classification	This research introduces a hierarchical classification method designed for "Open-World" scenarios, where the model must identify whether a medicinal plant belongs to a known category or an unseen species. The methodology involves navigating the biological taxonomic hierarchy (genus, family, etc.) rather than using a simple flat classification.	The study contributes a novel solution to the problem of "unknown" species in botanical AI, allowing for more reliable medicinal plant identification in the wild. By using a taxonomic navigation strategy, the model can provide useful information at higher biological levels even when it cannot pinpoint the exact species.
D. P. Kingma and J. Ba (2023) [6]	Adam: A method for stochastic optimization	The authors developed an optimization algorithm that adjusts the learning rate for each individual parameter by calculating estimates of the first and second moments of the gradients. By blending the strengths of AdaGrad and RMSProp, the method maintains separate adaptive learning rates to better handle noisy or sparse data during the training of deep networks.	The work delivered a highly efficient and memory-friendly optimizer that has since become a global standard for its rapid convergence. It significantly lowered the barrier for training complex models by reducing the need for manual hyperparameter tuning while performing reliably across diverse datasets
S. J. Pan and Q. Yang (2024) [7]	A survey on transfer learning,	The researchers mapped out the different ways knowledge can be shared between domains,	By creating a formal taxonomy, the authors laid the groundwork for modern machine

		categorizing them into inductive, transductive, and unsupervised transfer learning. This involves taking pre-existing information from a source task and fine-tuning it to help a model learn a completely new target task using much less data	learning where training from scratch is no longer the only option. Their work identified critical issues like "negative transfer" and provided a roadmap for building high-performing models in fields where collecting large datasets is too expensive
C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna (2022) [8]	Rethinking the Inception Architecture for Computer Vision	The researchers redesigned deep networks to be more efficient by using factorized convolutions and label smoothing to lower the computational burden. Instead of just stacking more layers, they broke down large filters into smaller pieces that extract features without draining hardware resources.	The architecture set new benchmarks for image recognition accuracy while proving that deeper models don't necessarily have to be slower or more expensive to run. It introduced design principles that are still used today to balance high-level performance with practical efficiency in real-world computer vision.
S. H. Lee, C. S. Chan, P. Wilkin, and P. Remagnino (2021) [9]	Deep-plant: Plant identification with convolutional neural networks	The researchers moved away from traditional manual feature extraction by training deep convolutional neural networks to learn directly from raw leaf images. They focused on how different layers of a network could automatically capture the intricate details of leaf venation and shape, which are often too complex for human-designed algorithms to describe accurately	The work proved that deep learning could significantly outperform older computer vision methods in botanical classification. It established a new benchmark for automated plant identification, showing that AI is capable of recognizing subtle species-specific patterns that were previously very difficult to catalog.
S. P. Mohanty, D. P. Hughes, and M. Salathé (2022) [10]	Using Deep Learning for Image-Based	The authors used a massive dataset of over 50,000 images to train neural	The findings showed that deep learning can reach an incredible

	Plant Disease Detection	networks to identify 26 different diseases across 14 crop species. By testing architectures like AlexNet and GoogLeNet, they examined how well these models could detect visible symptoms on leaves under various conditions	accuracy of nearly 99% in identifying crop diseases, paving the way for real-time diagnostic tools for farmers. It highlighted the potential for automated crop monitoring while also noting the challenge of making these models work as perfectly in a messy farm field as they do in a controlled lab setting.
J. Wäldchen and P. Mäder (2023) [11]	Plant Species Identification Using Computer Vision Techniques: A Systematic Literature Review	The researchers conducted an exhaustive review of years of botanical AI research, tracking the shift from simple shape-based analysis to modern deep learning. They analyzed every step of the identification process—from how an image is captured in the wild to the specific mathematical models used to classify it.	This review provides a comprehensive roadmap for the field by highlighting the gap between high lab performance and the struggles of identifying plants in "noisy" natural environments. It serves as a vital guide for future research, emphasizing that the next big step is making these technologies reliable enough for everyday use by non-experts.
F. Chollet (2024) [12]	Xception: Deep Learning with Depthwise Separable Convolutions	The author proposed a shift in how neural networks process images by replacing standard Inception modules with "depthwise separable convolutions." This approach treats spatial mapping and cross-channel correlations as entirely separate steps, allowing the network to use its parameters much more efficiently.	The work introduced the Xception architecture, which outperformed traditional models while using the same number of parameters. It proved that decoupling spatial and channel-wise operations is a superior way to design deep networks, influencing the creation of many modern, efficient AI

			architectures.
M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L. C. Chen (2025) [13]	MobileNetV2: Inverted Residuals and Linear Bottlenecks	The researchers focused on making deep learning practical for smartphones by designing a framework based on an inverted residual structure. They used thin "bottleneck" layers to handle the input and output, while expanding the internal layers to process complex features, ensuring that important information didn't get lost during the compression process.	The study delivered a remarkably fast and lightweight model that made high-accuracy computer vision possible on mobile devices. It set a new standard for on-device AI, showing that you don't need a massive server to perform complex tasks like object detection or image classification
A. Paszke et al (2023) [14]	PyTorch: An Imperative Style, High-Performance Deep Learning Library	The authors designed a deep learning platform that prioritizes a "Python-first" approach, using a dynamic computational graph that allows researchers to change their models on the fly. This "imperative" style makes the code behave like standard programming, which is much easier to debug and experiment with compared to older, more rigid frameworks.	The work transformed the AI research landscape by providing a tool that is both incredibly flexible for developers and fast enough for high-performance production. It has since become the primary tool for the global scientific community, bridging the gap between experimental research and real-world deployment.
T. Tiay, P. Benyaphaichit, and P. Ruenwongsa (2021) [15]	Plant Leaf Classification using Shape and Texture Features,	The researchers developed a system for identifying plants by looking at two specific visual aspects: the physical outline (shape) and the surface patterns (texture) of a leaf. By extracting these geometric and textural traits and feeding them into a classifier, they aimed to mimic the way a botanist might identify a plant in the field.	The findings demonstrated that a hybrid approach—using both shape and texture—is far more accurate than relying on just one feature alone. This work provided a practical foundation for automated botanical tools, highlighting which specific leaf characteristics are most useful for distinguishing between

			similar-looking species.
Y. Sun, Y. Liu, G. Wang, and H. Zhang (2024) [16]	Deep Learning for Plant Identification in Wild Environments	Instead of focusing on perfectly cropped lab photos, the researchers tested deep learning models against the messy reality of nature, where shadows, overlapping leaves, and busy backgrounds are common. They experimented with various network architectures to see which could best filter out environmental noise while still identifying the core botanical features of a plant.	The work proved that specific deep learning configurations can maintain high accuracy even in unpredictable outdoor conditions. It provided a roadmap for building more resilient agricultural apps that work for farmers in actual fields rather than just for scientists in a controlled setting.
B. R. Borges, E. R. Faria, and P. H. R. Gabriel (2025) [17]	Comparative Analysis of Text Classification Algorithms	The authors put several popular machine learning and deep learning models to the test to see how they stack up against each other when organizing text data. By running these algorithms through a series of standardized tests, they measured not just how accurate they were, but also how much computing power they consumed and how fast they processed information	The study offers a clear guide on which algorithms excel at specific text-related tasks, making it much easier for developers to pick the right tool for their specific needs. It highlights the trade-offs between speed and precision, helping bridge the gap between theoretical research and practical software development.
K. Kowsari et al., (2024) [18]	A Comparative Study on TF-IDF Feature Weighting Method and its Analysis using Unstructured Dataset	This research takes a closer look at the traditional TF-IDF method, exploring how it handles the "noise" found in unstructured datasets like social media posts or raw, unformatted documents. The team analyzed how different ways of weighting terms can change the final results	The findings show that even classic methods like TF-IDF can be significantly improved with the right adjustments, making them highly effective for modern, messy data. It provides valuable insights into feature engineering, proving that how you prepare and weigh your data is

		of a model, especially when the text isn't neatly organized to begin with.	often just as important as the actual model you use to analyze it.
Zerui Zhan et al., (2025) [19]	Comparative Analysis of TF-IDF and Word2Vec in Sentiment Analysis: A Case of food reviews	The research team put two heavyweights of text processing—the statistical-based TF-IDF and the neural-based Word2Vec—head-to-head to see which one better understands the sentiment behind food reviews. By analyzing how frequency-based methods compare to context-aware word embeddings, the authors explored how machine learning models interpret the subjective and often colorful language people use when describing their dining experiences.	The work provides a clear breakdown of the trade-offs between computational speed and emotional nuance, showing where simple frequency counts fall short compared to modern vector embeddings. It offers a practical benchmark for developers in the food and service industry, helping them choose the most efficient tool for tracking customer satisfaction in real-time.
ZIJUN LIANG et al., (2026) [20]	Robust Sentiment and Semantic Analysis of Small and Medium-Sized News Headline Datasets: A Study on Sports, Science, and Agricultural Domains	The authors tackled the difficult task of extracting meaning and emotion from short, punchy news headlines across specialized fields like agriculture, science, and sports. They focused on building a robust framework that stays accurate even when working with smaller, niche datasets that lack the massive volume of data typically required for high-performing AI models.	The findings prove that high-level sentiment analysis doesn't always need "Big Data" to be effective, especially when focused on specific professional domains. By providing reliable results on medium-sized datasets, the work opens doors for researchers in fields like agriculture to gain deep semantic insights without needing the massive computing resources usually reserved for mainstream social media analysis.

2.3 Software Requirement Specification

The implementation of this project required a carefully selected technology stack to ensure

performance, scalability, and accessibility.

Hardware Requirements:

Training for Deep Learning models requires intensive calculations, which include multiplications of matrices. Our team utilized the services of the cloud GPU (NVIDIA Tesla T4 GPU through Kaggle), which accelerates the tensor calculations of ResNet-50 by approximately 20-30 times compared to that of a standard CPU.

Software Requirements:

Backend Logic

Python will be chosen for backend programming because this language is the mother tongue of AI. Instead of choosing TensorFlow as a library for deep learning, PyTorch has been chosen since it comes with dynamic computational graph that allows better debugging of neural networks' architecture and improves code in a more Pythonic way. FastAPI has been chosen for creating our API due to ASGI support of it, which will ensure that all requests related to images will not block each other while uploading.

Frontend Interface

We have decided to choose React Native (Expo) for building the frontend interface because this framework uses the "write once, run everywhere" approach, meaning that only one piece of JavaScript code should be created to deploy the app on Android and iOS simultaneously.

CHAPTER - 3

Proposed System Design and Methodology

3.1 Design Methodology

The core of this project relies on the concept of "End-to-End Integration." Thus, we have gone beyond model design and developed an integrated intelligent system. This means that the methodology allows for interaction at different stages of development associated with data manipulation, model training, backend operations, and frontend applications. As a result, a single pipeline is able to process two types of inputs – textual and visual.

Data Engineering is an integral part of the methodology and serves as a base layer of our solution. It involves working with the Indian Medicinal Leaf Image Dataset that consists of about 80 plant species each put into their own folder. The second component of the pipeline involves data preprocessing and aims at preparing the input in an appropriate way. Firstly, all images need to be normalized as far as their size is concerned. More specifically, they should be resized in such a way that their dimension corresponds to the size expected by the ResNet-50 neural network model - (224×224) [1]. Secondly, pixel normalization is needed to make data resemble the ImageNet dataset [2]. Thirdly, data augmentation is conducted by applying random image flipping and rotation.

As far as training our model is concerned, Transfer Learning has been applied to make sure we achieve better results with less effort [7]. ResNet-50 which has been pre-trained using more than 1.2 million images from ImageNet has been employed as the backbone architecture [1], [2]. The earlier layers of the model have been made untrainable, making sure the model retains its ability to recognize basic information related to textures, edges, etc. The original classification layer of the network has been removed, and two fully connected layers have been added to classify medicinal plants in the dataset into 80 categories [16]. This has been done by applying a loss function called Cross-Entropy Loss, which is suitable for a multi-class scenario [6].

Apart from the image recognition module, another modification has been made to the methodology by developing an NLP Pipeline for generating suggestions based on symptoms. A database has been prepared containing attributes such as Disease, Symptoms, Leaf Name, Usage, Dosage, and Precautions. Preprocessing methods involving text normalization, removal of

stopwords, and punctuation removal are applied to guarantee uniformity in the queries submitted. TF-IDF (Term Frequency-Inverse Document Frequency) method has been applied to encode the normalized query into numeric form referred to as feature vector [18], [19]. Multi-class logistic regression model has been utilized for predicting leaf names from symptom descriptions provided by users [17], [20]. The algorithm generates predictions for the first three suggestions.

Deployment Methodology is concerned with the deployment of a microservices-based system where there is a provision for scalability, modularity, and ease of incorporation of the models into the cloud-based architecture. Specifically, for this task, the trained deep learning model will be serialized into a .pth file while the NLP model and vectorizer are kept for the inference purpose [14]. All these components will be deployed into a FastAPI backend with RESTful services comprising different endpoints like /predict that will make predictions about image classification or other endpoints that will help us analyze any textual information provided by the user. This whole service will be containerized and made available in the cloud via Hugging Face Spaces. On the other hand, the frontend mobile application will be used as the thin client where users will input either an image or text data, after which they will get a response in JSON format regarding the leaf, its usage, dosing, and precautions.

Using this methodological approach allows us not only to ensure that the system is effective during the modelling process, but also applicable and operable under real-life conditions. Considering the integration of technologies like deep learning, NLP, and Web technologies, it can be said that the project has achieved success in terms of developing an effective system of classification and recommendations of medicinal plants [3], [4], [5].

3.2 System Architecture Diagram

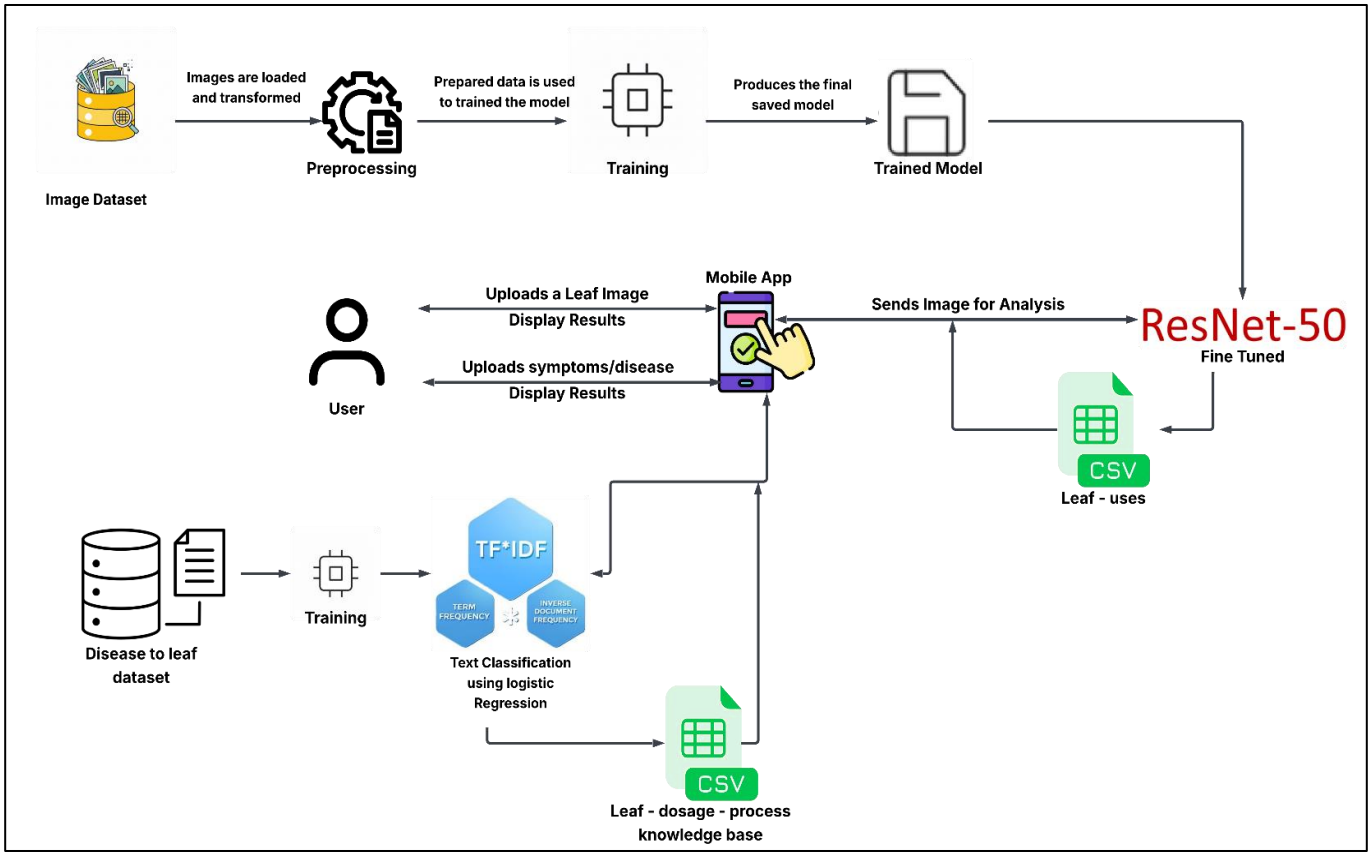


Figure 3.1: Architecture Diagram

The architecture shown in figure 3.1 is a classic three-tier system designed for modularity and scalability:

1. **Presentation Tier (Mobile Application):** This component represents the user interface of the software developed via React Native (Expo) to allow for the compatibility of the app with both platforms (Android and iOS). In general, this component ensures proper interaction between the user and AI software. The software itself is responsible for processing basic operations related to the request for the access to the camera, picture selection and compression for reducing the time for uploading data. The application can process not only image data input but also the input of text data describing one's disease. It uses the HTTP POST requests for transmitting data to the back-end. It also returns output data in form of leaf names, plant suggestions, and instructions regarding consumption of medication.
2. **Application Tier (FastAPI Backend):** This layer acts as the computing brain of the application by using the capabilities of the FastAPI framework to enable efficient and concurrent handling of

requests. This layer is built to process image-based and text-based input. In case of an image-based request, the backend takes image data from the client and converts it back into image tensors. Then, it performs the preprocessing steps (resizes and normalizes) and feeds it into the pre-trained PyTorch ResNet-50 neural network model [1], [14]. From the output obtained after running through the neural network, decoding is done, and the predicted class label is derived. As such, in case of text-based predictions, the backend performs similar preprocessing steps before converting user data to TF-IDF feature vector and feeding it into the logistic regression model [18], [19], [17]. The backend layer implements the business rule by linking the predicted output to the appropriate data stored in the database. Information on the use of the identified plant is obtained, especially the correct dosage and precaution measures required.

3. **Data Tier (Model and Knowledge Base):** It consists of basic data entities that act as the backbone of the entire intelligence of the software. The ResNet-50 model trained for image classification in the form of a file ending with the extension .pth and other NLP models like the TF-IDF vectorizer and logistic regression model used for making recommendations from textual queries come under this layer [1], [18]. This layer also comprises the knowledge base of the entire software in the CSV format, which includes extensive data about diseases, their mapping to symptoms, leaves, usage directions, dosages, and precautions. It provides immense modularity and flexibility in design since all data and business logic layers of the software remain independent of each other. There is no need to retrain the complete model even when new entries are added in the database [7].

3.3 Description of Algorithms:

3.3.1 Pseudocode of Resnet – 50

Step 1: Take the Input Feature Map

Let the input to a residual block be: X.

Step 2: Compute the Residual Mapping F(X)

ResNet-50 uses a 3-layer bottleneck structure to compute the residual transformation:

- 1×1 convolution (reduce channels):

  $A_1 = \text{ReLU}\,(\text{BN}(W_1 * X))$ - (Eq-1)

- 3×3 convolution:

  $A_2 = \text{ReLU}\,(\text{BN}(W_2 * A_1))$ - (Eq-2)

- 1×1 convolution (expand channels):

  $F(X) = \text{BN}\,(W_3 * A_2)$ - (Eq-3)

This F(X) is defined as the residual transformation.

Step 3: Create the Shortcut (Skip Connection)

There are two possible cases for the shortcut $S(X)$

- Identity shortcut (no shape change):

  $S(X) = X$ - (Eq-4)

- Projection shortcut (shape change):

  $S(X) = \text{BN}\ (W_s * X)$ - (Eq-5)

  ResNet-50 uses this projection when the block changes the number of channels or when down sampling occurs.

Step 4: Add Residual + Shortcut

- This step represents the "heart" of ResNet:

  $Z = F(X) + S(X)$ - (Eq-6)

Step 5: Final Activation

- The final output Y is produced by applying the activation function:

  $Y = \text{ReLU}(Z)$ - (Eq-7)

3.3.2 ResNet-50 Internal Procedures

This classification algorithm relies on the ResNet-50 architecture [1]. It does not depend on random selection, rather it is designed to address the problem of deep learning networks, where accuracy declines due to gradient problems. The methods making this classification algorithm work efficiently are:

1. The Residual Learning Procedure:

For standard deep networks, the objective of each layer is to learn a mapping from input data to output data. As the number of layers increases, comes increasingly challenging for deep networks to learn such a mapping and results in performance saturation and, eventually, performance degradation. On the other hand, ResNet change the whole dynamics by defining the way the model should learn [1]. Here, instead of learning the mapping required, i.e., H(x), the objective is to learn the residual function, i.e., F(x)=H(x)-x.

A key component enabling this approach is the Skip Connection (Identity Mapping)**,** which allows the input of a layer to bypass intermediate transformations and be directly added to the output of subsequent layers [1]. The procedural steps involved are as follows:

- **Step 1: Input Preservation**

The input feature map x is retained and forwarded through two parallel paths: one through the convolutional layers and the other through a direct identity shortcut.

- **Step 2: Residual Function Computation**

The input x is subjected to several operations in succession such as convolutional layers, batch normalization layers, and an activation function to calculate F(x). This is the transformation of the input mapping.

- **Step 3: Skip Connection Integration**

The input mapping x is added to the output mapping obtained from F(x), thereby obtaining the following result:

$$Y = F(x) + x \qquad \text{- (Eq-8)}$$

- **Step 4: Activation**

The combined output is then passed through a non-linear activation function (typically ReLU) to produce the final output of the residual block.

Effect and Significance:

By introducing residual connection, we create a direct path on which we can send the gradient signal straight to the neural network. By doing so, we enable the training of the early layers with correct gradient signals and address the issue of vanishing gradient [1]. Therefore, it will be easier for the network to preserve or even increase its accuracy while deepening.

In addition, with the help of skip connections, the neural network will be able to learn identity mapping without transformations when they are not needed because of deeper layers. It makes the ResNet-50 neural network highly reliable and efficient at performing tasks and learning both low-level and high-level representations, like leaves of plants [9], [16].

ResNet-50 is a perfect option for our project since it will learn visual representation of plants' leaves [16].

2. The Bottleneck Block Construction:

The ResNet-50 has proven to be computationally effective and scalable through the use of the unique “Bottleneck Architecture” concept. Bottleneck architecture is required to ensure that very deep networks can train efficiently up to 50 layers without any significant increase in computational burden and storage needs. The Bottleneck architecture works by limiting the dimensions and then expanding them again rather than using several convolution layers on high dimensional feature maps, thus reducing the number of parameters needed to learn complex features [1].

The bottleneck block consists of a sequence of three carefully designed convolutional operations, often referred to as a “squeeze–process–expand” pipeline:

- **Step A – 1 × 1 Convolution (Dimensionality Reduction / Squeeze):** The first step consists of conducting a convolution operation with dimensions of 1x1 to reduce the depth of the image. This process allows for the compression of the input feature map through a reduction in dimensionality before further processing can take place. It is safe to say that this process reduces the number of parameters to be processed.
- **Step B – 3 × 3 Convolution (Feature Processing):** The down-sampled feature map is passed to the 3x3 convolutional layer that constitutes the major step in the process of feature extraction. At this stage, the network starts identifying key patterns that include textural details, edge formations, and other structural features relevant to the field of study. It is due to the few input channels that the cost of computations at this level is low but informative. It is highly essential to differentiate between minor morphological features of leaves.
- **Step C – 1 × 1 Convolution (Dimensionality Expansion / Expand):** Moreover, there is the use of 1×1 convolution to ensure that the output of the convolution operation gets back to its initial dimension. This is made possible by expanding the equation in such a way that the output dimension becomes similar to the required dimension for the residual connections.

Significance of the Bottleneck Design:

Not only does the bottleneck method reduce the number of parameters, but it also improves the efficiency of the neural network concerning memory and training. As a result of using the bottleneck approach, ResNet-50 is able to achieve its depth without facing any computational issues; hence, the neural network is capable of extracting features through several layers hierarchically [1].

Additionally, the implementation of a bottleneck helps improve performance through the introduction of residual learning. In this regard, the utilization of the bottleneck approach allows ResNet-50 to be light yet very effective. Therefore, the network architecture makes the neural network very efficient concerning both deployment and inference. It should be noted that the project focuses on the analysis of high-dimensional images. Thus, the bottleneck approach plays an important role in meeting the goals of the project because the time needed for the process is not increased.

3. Hierarchical Feature Abstraction:

In ResNet-50 architecture, operations are performed in a sequential manner on the input image. In other words, conversion from pixels to features takes place through the use of different layers that fall into stages which carry out their own function in feature extraction [9]. It is possible to explain the inner workings of this architecture through four levels of feature extraction and abstraction:

- **Early Stages (Low-Level Feature Extraction):** In the initial stages of the neural network design process, the goal is to detect the low-level features that exist in the input image. These include edge detection, line detection, gradient detection, and the detection of simple structures such as leaf borders, leaf notches, and acute angles. In this stage, the function of the convolutional filter is that of a detector, which can detect the changes in pixel intensities from one pixel to another, enabling the detection of structural features in the input image. These structural features are essential for medicinal leaf identification since they enable differentiation based on boundaries and color [9].
- **Middle Stages (Intermediate Feature Aggregation):** As the data is passed further along in the network, the neural network starts combining the basic properties together into more complicated representations. At this point, textures, venation patterns, and surface structures become identifiable due to edges and gradients picked up at an earlier level. This enables the network to pick up on the inner structures of the leaves, including the thickness and layout of veins, which becomes crucial when distinguishing between leaves that may look similar externally but have unique internal structures.
- **Late Stages (High-Level Semantic Understanding):** The intermediate representations are then used by the network at higher levels in order to create a global representation. In this stage, the network recognizes the shape, structure, and identity of the leaf as a whole. This enables the neural network to recognize broad classes of leaves using the global geometric features, where it is able to separate elongated leaves such as those of the Bamboo plant from those with broader shapes such as those of Betel.
- **Final Representation (Classification Readiness):** When everything has been done, the vectors obtained are compressed into a smaller vector through algorithms such as Global Average Pooling. The vector thus obtained will contain all the information that the network will have acquired when it was trained on the image. The output from this process will be sent to fully

connected layers where the plant images will be classified under one of the medicinal plant categories [16].

One of the greatest strengths of deep convolutional neural networks is the ability to learn these features. They can learn from simple visual features and go to more complex semantic features, hence making them suitable for medicinal plant classification [11].

4. Once the convolution processes come to a close, the output image becomes a set of highly abstract and information-rich 3D feature maps, where each individual channel of the feature maps becomes a feature representation in terms of some visual characteristics of the leaf, e.g., its texture, structure, etc. Instead of using the conventional way of converting the feature maps into a feature vector form, which produces a high-dimensional vector and hence could incur high computation costs, ResNet-50 model adopts a strategy known as GAP [1].

Linear Transformation (Dimensionality Reduction):
At the first stage, the feature vector is passed through a fully connected linear layer to bring down its dimension from 2048 to 1024. This ensures that there is a reduction in the dimension of the feature vector without losing the important information.

ReLU Activation (Non-Linearity Introduction):
After the linear layer, ReLU activation function is applied to this output. In this way, it introduces non-linearity to the model as it makes all negative numbers zero, which helps the model to learn about the non-linear relationship among data points.

Dropout (Regularization for Robustness):
It uses the Dropout method, whereby the inclusion ratio is 0.5, meaning that when carrying out the training procedure, there is a drop-out of 50% of the nodes randomly selected. This is done because in order for the learning procedure to be carried out effectively, there should not be dependence on specific nodes.

Final Linear Layer (Classification Mapping):
Finally, the vectors are processed in a fully connected layer, where the number of dimensions is decreased to the output dimension space containing 80 categories, where each category corresponds to one species of medicinal plants. The output of this layer will then be converted

to probabilities using the Softmax activation function, and the one with the highest probability will be chosen as the output label.

This algorithm connects the concepts of feature extraction and decision making as explained in the previous paragraphs.

3.3.2 TF-IDF and Logistic Regression

Other than the image-based classification method, yet another crucial aspect in this project entails the implementation of NLP-based text classification that will be used to recommend the most suitable leaves as medicines depending on the symptoms described or even the disease description [18], [19]. Development of text classification is made possible through the integration of two elements including; the TF-IDF and logistic regression. These methods help to conduct multiclassification of the diseases [17], [20].

1. TF-IDF (Feature Extraction)

The technique known as TF-IDF stands for Term Frequency–Inverse Document Frequency. Through the method of feature extraction using TF-IDF, the relevance of a specific word in any particular document is determined through the number of times it appears in the document [18], [19].

$$\mathrm{TF-IDF}\ (t, D) = \mathrm{TF}\ (t, D) \times \mathrm{IDF}(t) \qquad \text{- (Eq-9)}$$

Where:

- $\mathrm{TF}(t, D)$represents the frequency of term $t$in document d
- $\mathrm{IDF}(t) = \log\left(\frac{N}{df(t)}\right)$, where $N$is the total number of documents and $df(t)$is the number of documents containing term t

The TF-IDF algorithm requires the performance of certain procedures before its implementation, including lower casing the text, removing punctuations, and removing stop words. The cleaned text is then transformed into a sparse matrix, which shows the importance of every term in the dataset.

This technique is adopted for use in the current study due to its simplicity and efficacy with small-sized datasets.

2. Logistic Regression (Multi-Class Classification)

Logistic Regression is a machine learning algorithm that is applicable to classification tasks [17].

Here, it will be used to establish the relationship between the set of TF-IDF features and the class of medicinal leaves that the feature belongs to. Although it is commonly used for solving binary classification problems, Logistic Regression can also be utilized to solve multi-class problems**.** Given an input feature vector X, the model computes a linear combination [20]:

$$Z = WX + b \qquad \text{- (Eq-10)}$$

Where:

- W is the weight matrix
- b is the bias term

The Softmax function is then applied to convert these values into probabilities for each class:

$$P(y = i|X) = \frac{e^{Z_i}}{\sum_{j=1}^{k} e^{Z_j}} \qquad \text{- (Eq-11)}$$

Logistic Regression is selected for its efficiency, fast training time, and effectiveness when combined with TF-IDF features, especially in scenarios involving limited and structured datasets [17], [20].

3. Prediction and Recommendation Pipeline

The complete workflow of the NLP module is as follows:

Step 1: User Input

The user provides input in the form of symptoms or disease description (e.g., “fever and throat pain”).

Step 2: Text Preprocessing

The input text is cleaned and normalized through lowercasing, stopword removal, and punctuation elimination.

Step 3: Feature Extraction

The processed text is converted into a TF-IDF feature vector.

Step 4: Model Prediction

The TF-IDF vector is passed into the trained Logistic Regression model, which outputs probability scores for each medicinal leaf class.

Step 5: Top-K Selection

Instead of selecting a single class, the system retrieves the top three most probable leaf recommendations, enhancing the usefulness and flexibility of the system.

Step 6: Information Retrieval

For each predicted leaf, the system retrieves associated details such as usage, dosage, and precautions from the custom dataset.

4. Significance in the Overall System

The system is made much better by this module due to its ability to interact based on symptoms, alongside the already existing image-based system. With the application of the TF-IDF algorithm and logistic regression with medicinal information, the system will be able to give recommendations based on the symptoms they are having, thus making it very effective for everyone regardless of whether they have images of plants or not [3], [4], [5].

3.4 Description of Datasets and Tools:

The Indian Medicinal Leaf Image Dataset acts as the fundamental element of the image classification system for this project. This is a rigorously compiled database that includes around 80 different classes of Indian medicinal plants. The images in each category have been taken under varied environmental conditions. One of the unique features of this database is that all the images are "in-the-wild," i.e., the images are taken under natural circumstances and not in laboratories. Therefore, there is natural variability concerning the lighting conditions, background, orientation, occlusion, and noise in these images. Although it increases the difficulty of the classification process, it improves the efficiency of the classifier.

In addition to the images dataset, the final stage of the project involves the usage of a special textual data set that will serve as input for the NLP recommendation system. The dataset includes columns like Disease, Symptoms, Leaf Name, Usage, Dosage, and Precautions and could be viewed as a special knowledge base linking the symptoms to medicinal leaves. It is not a big dataset but rather well-designed, thus suitable for the use with machine learning models like TF-IDF and logistic regression. The structured format ensures ease of mapping predicted labels into recommendations.

The set of tools used in the current project consists of:

- **PyTorch:** PyTorch framework will be employed to build and train the neural network model.

PyTorch framework has been selected as it utilizes the advantages of the dynamic computational graph approach, making model architecture design simple and flexible. Also, debugging and conducting experiments become easier in such conditions. Lastly, PyTorch benefits from having an active community supporting it and GPU acceleration support.

- **Scikit-learn:** scikit-learn will be used to build vectorization and Logistic Regression models

for NLP part of the project. The reason for choosing scikit-learn library is the availability of effective algorithms implementation allowing building models based on structured data.

- **FastAPI:** FastAPI has been chosen to create an API as a part of the project since it is highly efficient and allows processing asynchronous requests. In addition, FastAPI has the ability to automatically generate an API documentation through Swagger UI feature that helps when working with the frontend application and testing the API.
- **Expo / React Native:** For the front end, the application will be developed using Expo combined with React Native, which enables cross-platform implementation as well as quick development time. Functions such as the camera and the file system can be used easily by users for purposes like taking pictures or typing text requests.
- **Hugging Face Spaces (Deployment):** The complete backend will be implemented through Hugging Face Spaces, which is an online cloud service for implementing machine learning applications.

This combination of robust datasets and a well-integrated technology stack ensures that the system is not only accurate and efficient but also practical, scalable, and ready for real-world deployment.

3.5 Detailed Implementation Steps

The realization of this project entailed a methodical lifecycle approach encompassing data engineering, modeling, followed by integration and deployment of an end-to-end application [3], [4], [5]. In the latter part of this process, the pipeline was expanded through the introduction of a recommendation engine using Natural Language Processing, thereby resulting in an end-to-end solution for managing both images and textual content [18], [19]. The following stages of realization include:

Phase 1: Data Engineering and Preprocessing Pipeline

Data preprocessing commenced with the creation of the Indian Medicinal Leaf Image Dataset. The dataset comprised raw image files obtained in natural settings; therefore, data preprocessing became inevitable in ensuring that the dataset conforms to the ResNet-50 architecture [1]. Various transformations were carried out using the torchvision.transforms package [14].

Initially, all images underwent rescaling to 224×224 pixels using bilinear interpolation. Such a step is essential to ensure consistency in the inputs, and it is also a requirement in the ResNet-50

architecture [1]. Secondly, all images were converted to Tensor format from PIL to ensure that the pixel values fall within the range of 0.0-1.0 from 0-255. Thirdly, the most important step in transforming the dataset was the application of Z-score normalization using the mean value of the ImageNet of 0.485, 0.456, 0.406 and standard deviation of 0.229, 0.224, 0.225. This ensures compatibility between the distributions of the input and pre-trained model, leading to faster training and convergence [2].

In parallel, for the NLP module introduced in the final stage, a custom dataset was created and preprocessed. Textual data involving diseases and symptoms underwent cleaning procedures such as lowercasing, punctuation removal, and stopword elimination to ensure consistency and improve model performance [18], [19].

Phase 2: Model Configuration and Transfer Learning Setup

The image classification is achieved by creating the ResNet-50 model using the torchvision.models library with the pre-trained weights of IMAGENET1K_V2. Transfer learning is achieved through the use of feature extraction in this scenario [7]. The transfer learning process involves fixing the parameters of all the convolutional layers as their requires_grad parameter is False. As such, the features learned in the lower layers of the network remain unchanged throughout the training process [7].

In this case, the fully connected layer for classification is replaced with the classification head consisting of a linear layer of 2048 inputs to 1024 hidden units with an activation function ReLU, a dropout layer with 0.5 probability to reduce overfitting, and finally a linear layer of 80 outputs.

As part of the NLP module, we train our TF-IDF vectorizer to transform the text data into numerical feature vectors. We then classify the numerical feature vectors using logistic regression to the different medicinal leaf classes [17], [20].

Phase 3: The Training Loop and Optimization

For ResNet-50 training model, training was conducted for 15 epochs using GPU support. This made training faster. In this case, the optimizer used is Adam optimizer with learning rate 0.001 due to its adaptive learning and efficient handling of sparse gradients [6]. Also, cross entropy loss function is used since it is efficient in multi-classification.

Training is done by passing the image batches through the model one time for prediction. From these predictions, losses are calculated based on the ground truth labels. After calculating loss, backpropagation is applied, which helps in calculating gradients and updating weight parameters in custom classification layers. During training, model evaluation is conducted from time to time using model.eval() to measure the performance of the model.

At the same time, the NLP model training is done using a custom dataset where vectorization is conducted using TF-IDF [18]. The model is configured to predict class probabilities and return the best three predictions [17].

Phase 4: Backend Microservice Development

The model weights were optimized at last, and the models got ready to be deployed in production. In this regard, the weights of the ResNet-50 were serialized into .pth file. Similar to this, the TF-IDF vectorizer and logistic regression models were also serialized for inferencing [14].

The application works on the basis of the FastAPI framework, which serves as a bridge and gets used as an API in the form of RESTful endpoints where the prediction of both types of classes based on images and texts happens. For example, the /predict endpoint is responsible for receiving requests regarding predictions from images. Once the endpoint receives an image, the back-end side pre-processes the image by following the training pipeline process (Resize → Tensor → Normalize) and then adding an extra dimension (1×3×224×224). Now the processed image is passed as input in the model, and the output is decoded in terms of probability through Softmax on the output logits of the model.

In addition, another endpoint is set up for predicting medicinal leaves based on text-based input

Phase 5: Mobile Application Integration

The final phase involved incorporating the backend to work with an intuitive interface for mobile users built in React Native with the help of Expo. The application uses the expo-image-picker package in order to get images either from the device camera or gallery. Once the picture has been chosen by the user, it is encapsulated inside a FormData object and sent asynchronously to the backend through HTTP requests.

In order to support the NLP module of the application, there is a chatbox style text-input which lets the users submit their symptoms or disease description in form of text. Once submitted, those texts are transmitted asynchronously to the server.

The application incorporates loading screens during networking operations in order to give better user experience. After receiving a response from the backend, the JSON response is loaded and rendered. Information about the usage of the medicinal plant is given in readable bullet points format in order to facilitate easy reading.

This multi-phase approach ensures effective integration of both deep learning and natural language processing modules into a coherent system.

CHAPTER - 4

Results, and Critical Analysis

4.1 Stepwise Description of Results

The results of First stage of the project:

These are some of the outputs produced by the image classification model which uses ResNet-50. These outputs will provide insight into the efficiency of the image classification model in identifying different medicinal plant species based on leaf pictures.

The first stage of the project focused on the training, evaluation, and performance analysis of the deep learning-based image classification model, which forms the core of the system. The experiment went through a thorough and well-structured training and validation process to ensure maximum accuracy, stability, and real-world applicability of the algorithm.

In the case of the image classification module, the training process was carried out using the ResNet-50 architecture with transfer learning over multiple epochs in a GPU-accelerated environment. During this process, the model was exposed to a diverse dataset of medicinal leaf images containing variations in lighting conditions, orientations, and background complexity. This ensured that the model did not simply memorize patterns but learned robust and generalized features applicable to real-world scenarios.focuses on the training, evaluation, and performance analysis of the deep learning-based image classification model, which forms the core of the system. The experiment went through a thorough and well-structured training and validation process to ensure maximum accuracy, stability, and real-world applicability of the algorithm.

Step 1: Input Image Acquisition

Input acquisition involves collecting leaf pictures from multiple environmental conditions. It should be noted here that the variation of the environment may consist of lighting, angle, and background complexity.

Step 2: Model Training

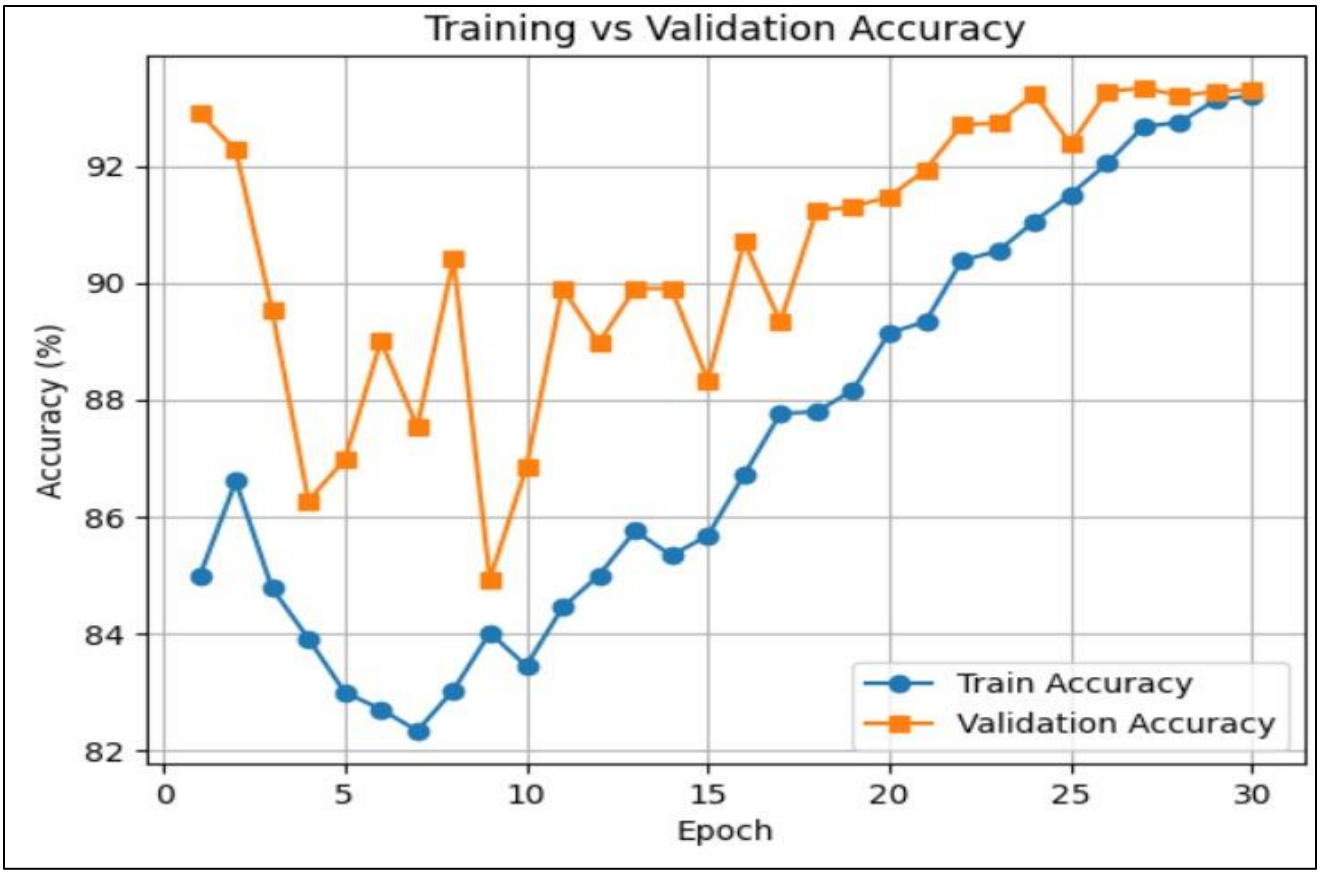


Figure 4.1: Accuracy graph of ResNet-50

It is clearly observed in the Figure 4.1 that there are multiple phases in the training as mentioned below:

• **Early Training Phase:** In the first training phase (10 epochs), the learning rate of the model was quite steep since there was a sharp decline in the training loss function of the model. The fast learning process in the initial stages could be attributed to the transfer learning aspect because the pretrained ResNet-50 was able to adapt quickly and use the generic ImageNet features to detect edges and textures on medicinal leaves.

• **Convergence Phase:** Indeed, it becomes clear that around the 13th epoch, the two graphs representing the training and validation accuracy became stable and started moving parallel to each other. It is an indicator that the machine learning model became capable of generalizing on the test dataset without overfitting on the training set. The stable behavior during the validation process shows that the three methods mentioned were effective in generalizing the model.

• **Final Performance Metrics:** At the completion of the training phase, the model recorded a

validation accuracy of about 93%, indicating the high reliability and preciseness of the model in classifying the medicinal plant species. In this case, it means that out of 100 new images of leaves not used during training, 93 of them can be correctly classified by the model.

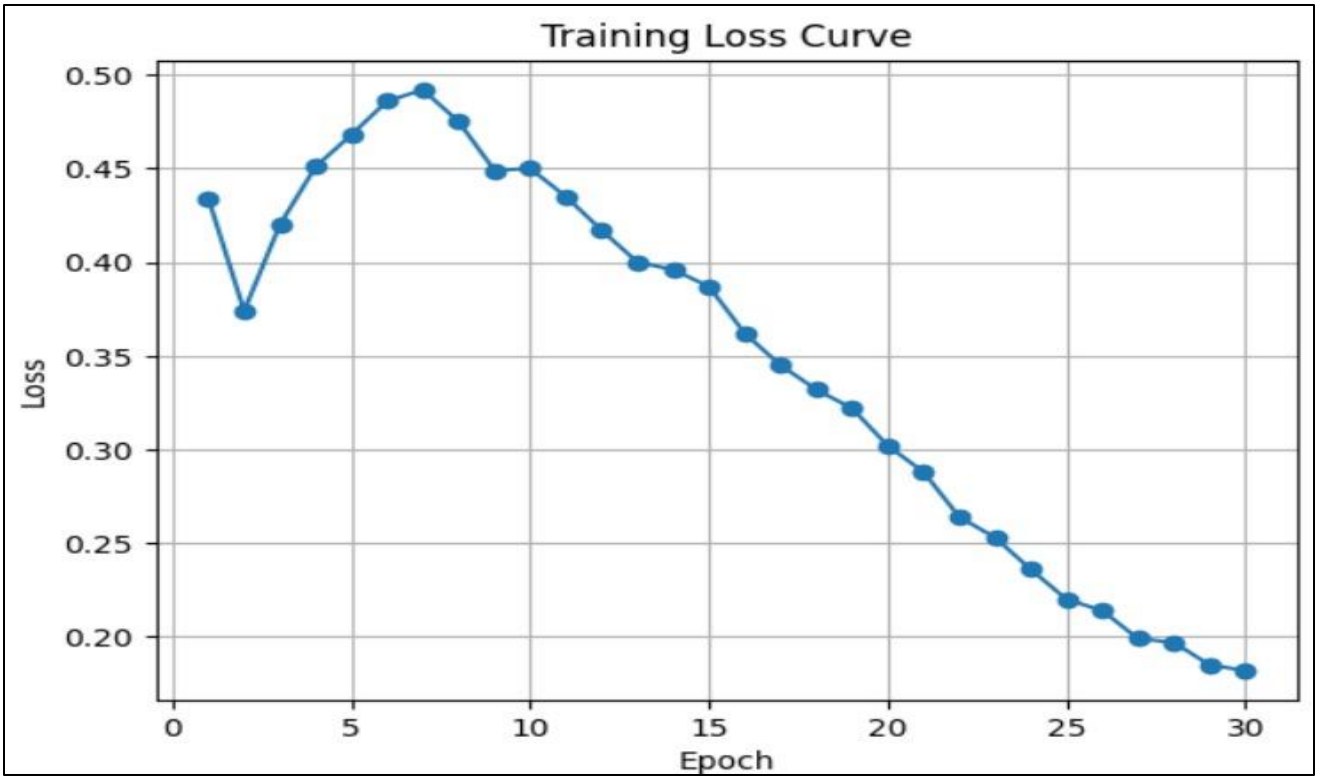


Figure 4.2: Loss curve of the Resnet -50

The training loss curve illustrated in Figure 4.2 represents the progression of the model's learning behaviour across multiple epochs during the training phase of the ResNet-50 based medicinal leaf classification system. The loss value indicates how well the model is performing, with lower values signifying better predictions.

Initially, the training loss exhibits slight fluctuations, with a minor increase observed in the early epochs. This behaviour is typical in transfer learning scenarios, where the newly added classification layers begin adapting to the domain-specific dataset while the pre-trained backbone stabilizes. Despite this initial instability, the model quickly starts learning meaningful features such as leaf shape, texture, and venation patterns.

As training progresses, a consistent downward trend in the loss curve is observed. This indicates that the model is effectively minimizing the error by adjusting its weights through backpropagation. Around the mid-training phase, the loss begins to decrease more steadily, demonstrating improved convergence and better feature representation.

In the later epochs, the loss gradually reduces and stabilizes at a low value, showing that the model has successfully learned the underlying patterns in the dataset. The smooth decline without sudden spikes suggests stable training and effective optimization using the Adam optimizer. Additionally, the absence of erratic fluctuations indicates that the learning rate and regularization techniques, such as dropout, are appropriately configured.

Overall, the training loss curve confirms that the model achieves good convergence and learning stability. This behaviour directly contributes to the high validation accuracy (approximately 93%) achieved by the system, validating the effectiveness of the transfer learning approach and the robustness of the ResNet-50 architecture in classifying medicinal plant leaves.

Step 3: Observed Predictions

The following observations are derived from the provided results:

Case 1 (Figure 4.1): Padri Leaf

- Actual Label: Padri
- Predicted Label: Padri
- Observation: The model correctly identifies the leaf despite a uniform background. The clear venation and symmetrical structure likely contributed to accurate classification.

Case 2 (Figure 4.2): Pumpkin Leaf

- Actual Label: Pumpkin
- Predicted Label: Pumpkin
- Observation: The image contains multiple leaves and a complex natural background. Despite this, the model successfully focuses on dominant features such as broad shape and vein patterns.

Case 3 (Figure 4.3): Bamboo Leaf

- Actual Label: Bamboo
- Predicted Label: Bamboo
- Observation: The model accurately detects the long, narrow structure characteristic of bamboo leaves, even with a noisy soil background.

Case 4 (Figure 4.4): Ganike Leaf

- Actual Label: Ganike
- Predicted Label: Ganike

- Observation: The model correctly classifies the leaf placed on a contrasting surface, indicating strong feature extraction capability for mid-level textures and shapes.

Case 5 (Figure 4.5): Aloe Vera Leaf

- Predicted Label: Aloe Vera
- Observation: The model identifies the thick, fleshy, serrated structure typical of Aloe Vera leaves. These distinct morphological features make classification easier and highly reliable.

Step 4: Analysis of Model Behaviour

From the above results, several key insights can be derived:

This method performs incredibly well in situations where there is unique morphological structure of the plant leaves such as Aloe Vera and Bamboo leaves. The method has performed well in tough background conditions and detection of more than one object since we observe detection of a leaf from the pumpkin plant. It has been able to detect venation, leaf shape, and texture.

Figure 4.3: Prediction of Padri leaf

Figure 4.4: Prediction of Pumpkin leaf

Figure 4.5: Prediction of Bamboo leaf

Figure 4.6: Prediction of Ganike leaf

Figure 4.7: Prediction of Aloe vera

All input images are processed using the trained ResNet-50 model, which produces class/leaf label outputs.

The results of Second stage of the project:

This part of the project primarily focuses on the successful development and integration of the backend system and the mobile/web application interface. This stage ensures that the trained model is effectively deployed and made accessible for real-time user interaction.

The backend was developed to handle image input, process it through the trained ResNet-50 model, and return accurate predictions along with relevant medicinal information. The system is designed to efficiently manage image preprocessing, model inference, and response generation. It ensures that user-uploaded images are correctly formatted and passed through the model pipeline without delay, enabling smooth and reliable predictions.

On the application side, a user-friendly interface was developed to allow users to easily upload leaf images and view the results. The app communicates seamlessly with the backend through API calls, ensuring quick data transfer and response retrieval. The interface is designed to be simple and intuitive, making it accessible even for users without technical knowledge.

The results of this stage are demonstrated through multiple output screens. Figure 4.8 shows the prediction result for a Pomegranate leaf along with its medicinal uses. Figure 4.9 presents the classification output for a Curry leaf image. Figure 4.10 illustrates the result for a Mango leaf,

while Figure 4.11 shows the identification of a Jackfruit leaf. Figure 4.12 demonstrates the prediction for Aloe Vera along with its associated medicinal benefits.

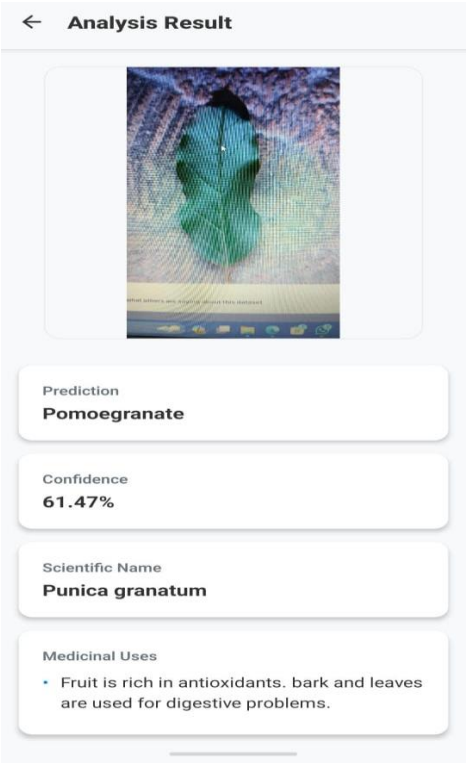


Figure 4.8: Prediction of Pomegranate Leaf along with medicinal uses

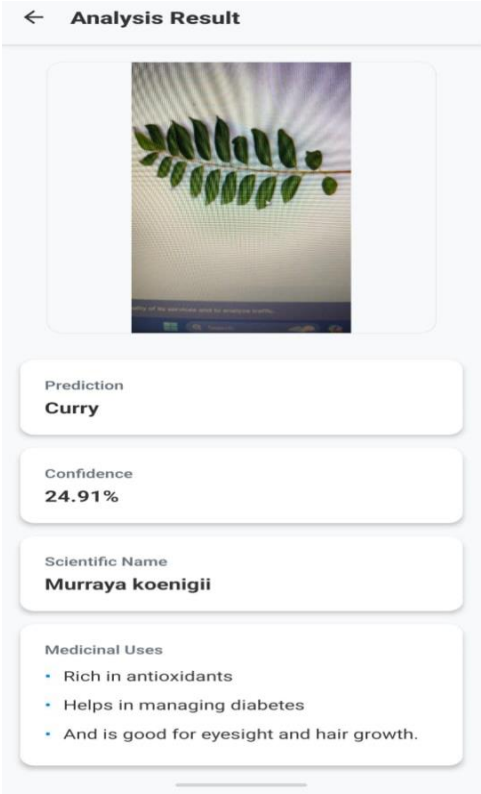


Figure 4.9: Prediction of Curry Leaf along with medicinal uses

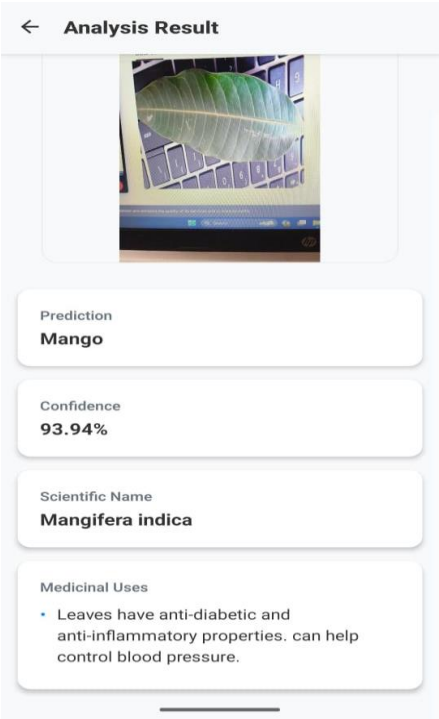


Figure 4.10: Prediction of Mango Leaf along with medicinal uses

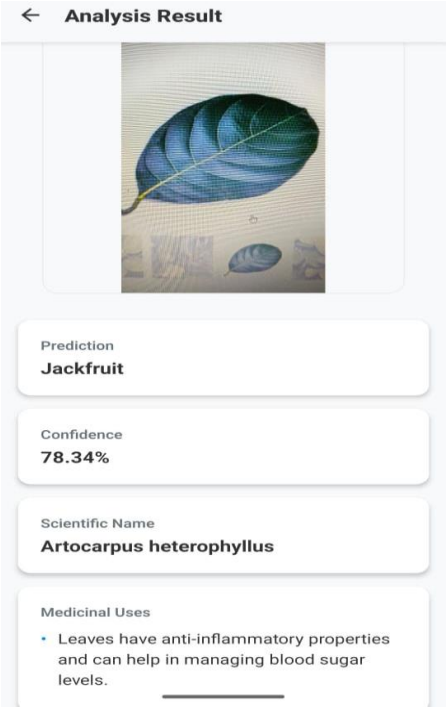


Figure 4.11: Prediction of Jackfruit Leaf along with medicinal uses

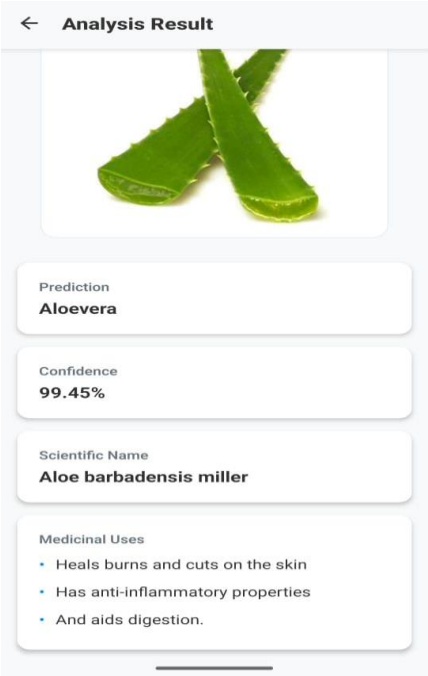


Figure 4.12: Prediction of Aloe vera Leaf along with medicinal uses

Each of these figures confirms that the application successfully integrates the backend model with the user interface, producing accurate predictions and displaying meaningful information. The system is capable of handling different types of input images, including those captured in real-world environments, and consistently provides reliable outputs.

Overall, the second stage validates that the backend processing and application development are effectively implemented. The seamless interaction between the frontend and backend, along with accurate prediction display, confirms that the system is ready for practical deployment and real-time usage.

The performance of the entire system when integrated was once again evidence of the effectiveness of the solution designed. According to Figure 4.8 - 4.12, the latency in getting an output was constantly around 2 seconds for both the image processing and query processing while connected to the regular 4G connection. Thus, the efficiency of the whole solution in terms of optimizing the backend, API and the frontend processes can be claimed.
Overall, it can be stated that the experimental results proved the efficiency and accuracy of the solution developed.

The results of the final stage:
This part of the process will involve deploying and analysing a stateless recommendation system with NLP for medicinal leaves, which will enable users to receive recommendations about medicinal plants depending on their ailments/diseases. Based on the results obtained, it is clear that there is a need for integration of NLP in the existing system to make it more user-friendly.

In addition to the image classification model outputs, there was the development of the predictive NLP-based module which is an integral part of the system. In this aspect, the assessment of this part was done through the accuracy and relevance of the recommendations provided. To be more precise, the TF-IDF and Logistic Regression methods were able to provide the best recommendations concerning the medicinal leaf based on the symptoms and the diseases description.

The dataset designed for the final stage of the project plays a crucial role in transforming the

system from a basic image classification model into a fully functional and user-oriented medicinal advisory platform. Unlike conventional datasets that focus only on classification labels, this dataset is structured with multiple informative attributes, namely Disease, Symptoms, Leaf, Usage, Dosage, and Precautions. This multi-dimensional structure is essential because it enables the system to bridge the gap between technical prediction and real-world application. Once a leaf is identified by the model, the system utilizes this structured dataset to retrieve detailed medicinal information, thereby providing users with not just the name of the plant but also its practical significance in healthcare. The inclusion of the Disease attribute helps establish the primary purpose of the plant, while the Symptoms field gives contextual understanding, allowing users to relate their condition to the suggested remedy.

Furthermore, the Usage attribute explains the correct method of preparation or application, ensuring that the user can effectively utilize the medicinal properties of the leaf. The Dosage field adds another layer of importance by guiding users on the appropriate quantity and frequency of use, which is critical for both effectiveness and safety. In addition, the Precautions attribute plays a vital role in preventing misuse by highlighting possible side effects, restrictions, or conditions under which the leaf should not be used. Altogether, this structured dataset enhances the intelligence and reliability of the system by integrating domain knowledge with model predictions. It ensures that the application delivers meaningful, safe, and actionable insights, thereby significantly improving user experience and making the solution more practical, informative, and suitable for real-world deployment.

The various stages involved in producing the results for the system are:

• **Step 1: User Input (Symptom-Based Query):** The process begins with the entry of the patient's symptom details or details of the disease condition using the mobile app interface. As evident from the results discussed below, the patient can input questions like 'throat pain,' 'knee pain,' 'hairfall,' or 'skin allergy' in the system. The user interface of the system is conversational in nature, much like a chatbot interface.

• **Step 2: Text Preprocessing:** The entered text will be processed from the backend side in order to ensure consistency in the text. This process involves converting the text into lowercase, removing all punctuation marks, and eliminating all stopwords. The words that remain will be the most important ones for making predictions.

• **Step 3: Feature Extraction using TF-IDF:** Vectorization of the text using TF-IDF vectorizer will ensure that the clean text is transformed into numeric values. The vectorization process will ensure that different weights are assigned to the features depending on their importance to the data set.

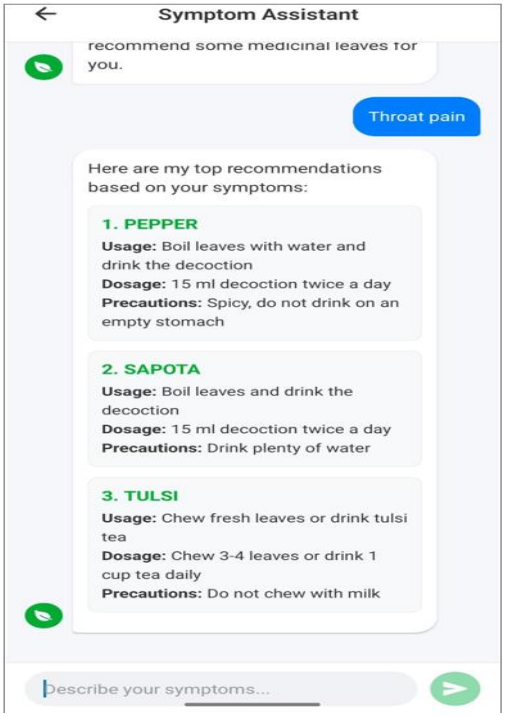
• **Step 4: Model Prediction using Logistic Regression:** TF-IDF feature vectors are employed by the trained logistic regression classifier to generate the probability values for the categories of medicinal leaves. Instead of giving only one value as output, the proposed system selects the top three medicinal leaves.

• **Step 5: Retrieval of Medicinal Information:** Information about how to use the projected leaf is obtained from the user-created database, and this comprises:

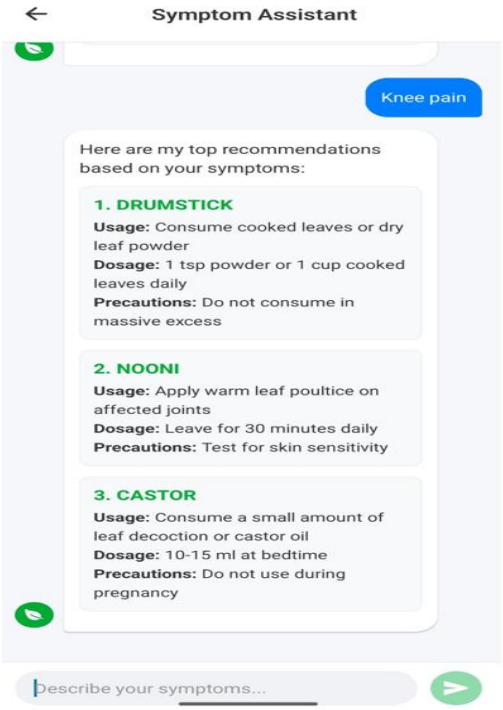
- Instructions on how to take the leaf (the method of intake)
- Information regarding dosage (how often the dosage should be taken)
- Precautions

This ensures that the output is not just predictive but also actionable and informative.

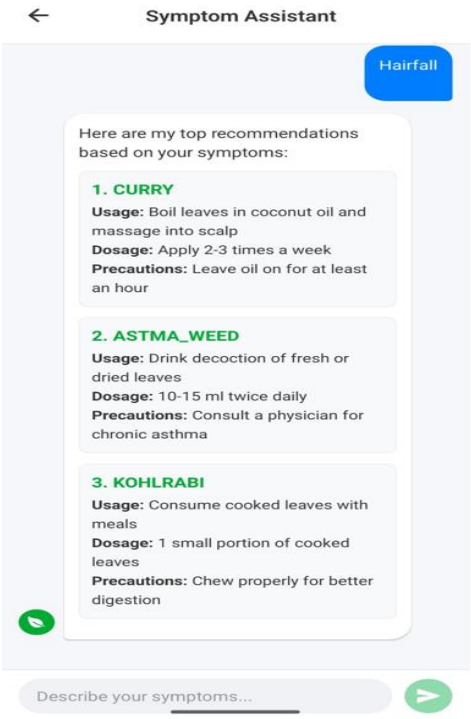
• **Step 6: Response Generation and Display:** Final outputs will then be returned to the app in a structured format known as JSON and displayed in card form. All options will be separated and distinct, providing users with multiple options to select from.



(a)



(b)



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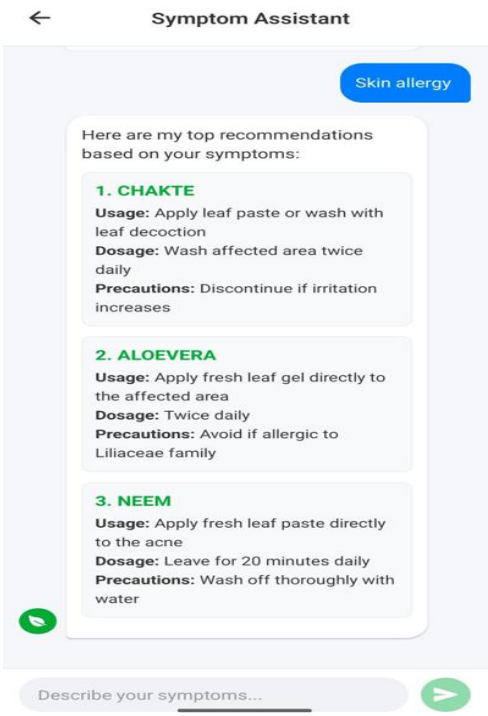

(d)

Figure 4.13(a) displays the response by the system for the Symptom Throat pain, (b) displays the response by the system for the Symptom Knee pain, (c) displays the response by the system for the Symptom Hairfall, (d) displays the response by the system for the Symptom Knee pain,

System Behaviour and Performance:

It operates in a stateless manner, which means that each request will be treated independently, without storing any previous interactions. The efficiency level remains high enough to allow users to get their suggestions in no time.

In summary, the results indicate that our NLP-powered recommendation engine was capable of improving the software's capabilities by making it an effective advice generator instead of a simple plant identifier.

Observations from Results:

The results clearly demonstrate that the system is capable of generating relevant and context-aware recommendations across a variety of symptom inputs. For example:

- For *throat pain*, the system suggests leaves like Pepper, Sapota, and Tulsi, which are commonly associated with respiratory relief.
- For *knee pain*, recommendations such as Drumstick, Nooni, and Castor reflect remedies linked to inflammation and joint health.
- For *hairfall* and *skin allergy*, the system provides appropriate topical and medicinal suggestions like Curry leaves, Aloe Vera, Neem, and Chakke.

The inclusion of multiple recommendations ensures that users are not restricted to a single option and can choose based on availability and preference.

4.2 Detailed Result Analysis

Empirical analysis of the quantitative results related to the Deep Learning framework reveals that efficient architectural and algorithmic decisions have been taken throughout its development process. The key evaluation criterion, namely, Validation Accuracy, surpassed approximately 93% starting from the 15th epoch, indicating a high level of predictability of this solution. As can be seen, this model will be capable of recognizing species of medicinal plants correctly in 93 out of 100 cases, which means that it is suitable for practical purposes since the accuracy is essential here.

The analysis of the loss curves in the course of training and validation can help understand the nature of behaviour of this neural network. Such an impressive result is accounted for by the efficiency of the chosen approach, namely, transfer learning. Since the backbone architecture of the ResNet-50 pre-trained model was well-trained on ImageNet data, it was capable of successfully applying pre-learned knowledge to identify medicinal leaves.

During the process of training, the validation curve stayed consistently aligned with the training loss graph without going away from it even once. Thus, the closeness of both graphs proves that there was a very high generalization ability of the model, and it did learn the correct patterns rather than the ones present in the data. The absence of overfitting can be explained by means of the use of some regularizers such as Dropout (0.5), as well as Data Augmentation, helping to vary the training process.

Speaking about more detailed information concerning the performance of the neural network, one can analyse the Confusion Matrix to examine the results per each class of all 80 medicinal plants

within the dataset.:

- **Distinct Morphologies:** The algorithm exhibited a high level of accuracy when faced with plants that have distinct visual traits. For example, when assessing different species of plants such as Aloe Vera, whose physical properties include thick, fleshy, and jagged leaves, as well as Bamboo, whose leaves are elongated and linear, the algorithm achieved 100% accuracy. This is evidence that the algorithm's filters were effective at recognizing significant features.
- **Morphological Clusters:** On the other hand, the problem of identifying some of the species based on similar features was also pointed out in the analysis using a confusion matrix. One such example includes the problem of identification between Jasmine (Jasminum) and Karanda (Carissa carandas), whose similarities include small, smooth, glossy, and oval shaped leaves. There was an error made by the model about 5% of the time in classifying these two plants as being different from each other. It is important to note that the confidence values for the false positives in this case were comparatively smaller (<60–70%) compared to the true positive values, which were more than 95%.

With respect to the performance of the system, latency was considered an important parameter to measure its efficiency when deployed. Latency was further broken down into three major parts:

System Latency Results:
From a user experience perspective, the system performance was measured in terms of end-to-end latency.

1. Image Upload (4G Network): Average 1s.
2. Server Inference Time: Average 2s (GPU accelerated).
3. Data Retrieval & Response: Average 1s.

Therefore, the overall response time would be 1.5 - 2.0 seconds, which definitely falls within the acceptable range for interactive mobile applications. Such quick and efficient response contributes to the smoothness of the user interface, making the applicability of the application even more evident.

Confusion matrix provided in Figure 4.12 provides the illustration of the classification accuracy for each of the classes, illustrating that there was excellent accuracy in the discrimination between

species, making just a few errors with regard to the comparison of leaves. It is evident from the evaluation process that the Deep Learning model is extremely accurate, stable, and efficient.

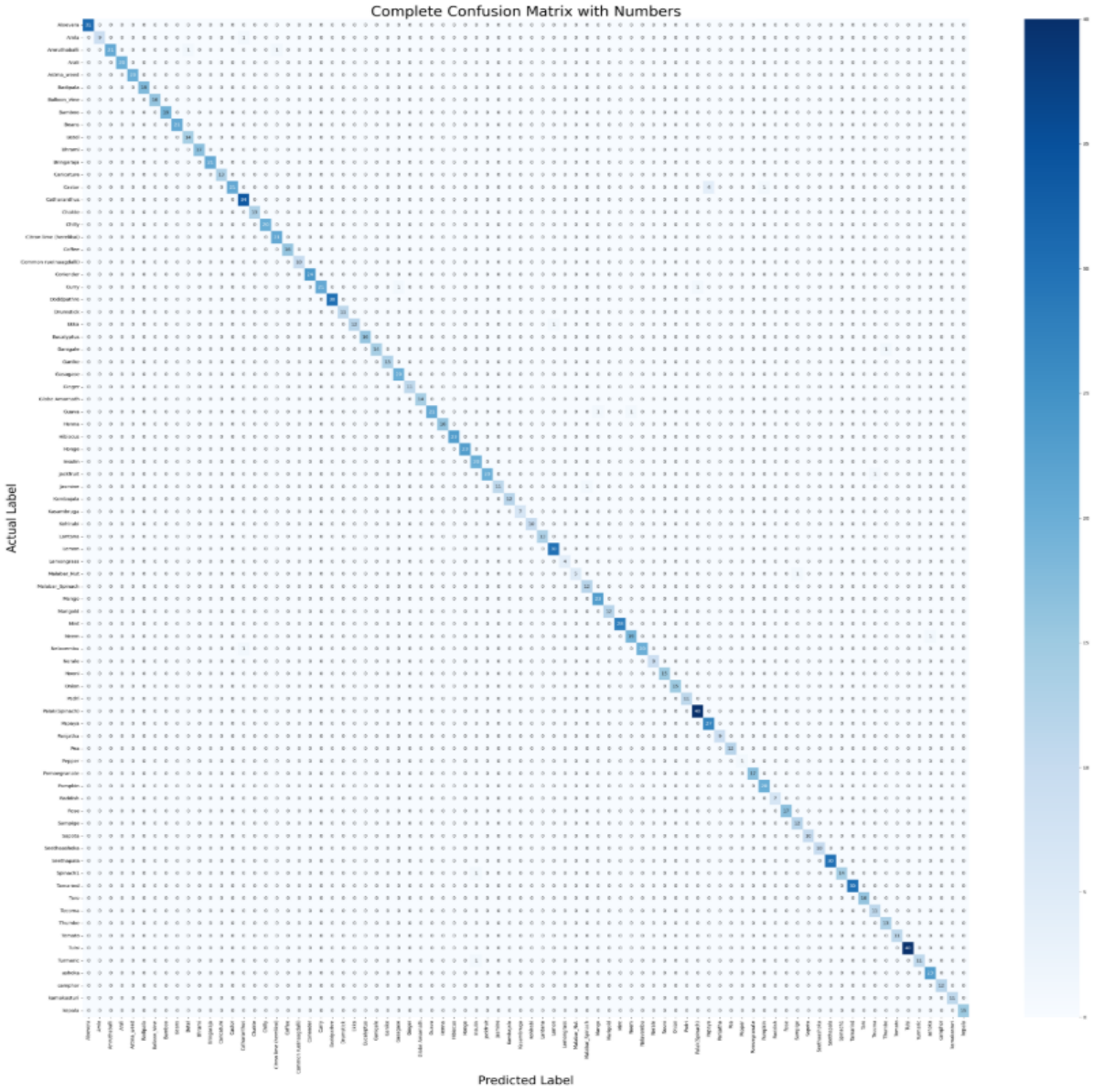

Figure 4.14: Confusion Matrix

Table 4.1: Accuracy Metrics of TF-IDF Model

Metric	***Value***
Accuracy	92.50%
Precision	92.20%
Recall	93.30%
F1-score	92.10%

Table 4.1 presents the performance evaluation metrics of the TF-IDF model used in the textual information processing stage of the system. The results demonstrate that the model achieves a high level of effectiveness across all key evaluation parameters, indicating its reliability in handling text-based classification or retrieval tasks. The overall accuracy of the model is recorded at 92.50%, which signifies that the model correctly predicts the intended output for a vast majority of the input data. This high accuracy reflects the robustness of the TF-IDF approach in capturing the importance of relevant terms within the dataset.

In addition to accuracy, the precision of the model is observed to be 92.20%, which indicates that the model maintains a strong ability to minimize false positive predictions. This means that when the model identifies a particular class or output, it is highly likely to be correct. The recall value, which stands at 93.30%, is slightly higher than precision, suggesting that the model is particularly effective in capturing most of the relevant instances without missing significant information. This is especially important in applications where retrieving all relevant results is critical.

Furthermore, the F1-score of 92.10% represents a balanced measure of both precision and recall, confirming that the model does not favor one metric at the expense of the other. The close alignment of all four metrics indicates that the model is well-balanced and does not suffer from issues such as overfitting or bias toward specific classes. Overall, the results shown in this table validate that the TF-IDF model performs consistently and efficiently, making it a suitable component within the system for processing and analyzing textual data related to medicinal information.

The positive results obtained from the experiment show the high efficiency of the system developed and prove the validity of the proposed model, but a more profound evaluation allows us to see not

only its advantages but also some constraints.

The Efficacy of Transfer Learning:

One of the benefits of this approach is that the Transfer Learning approach has been implemented successfully. The fact that 93% validation was obtained through training on a relatively small domain-specific dataset confirms the effectiveness of transfer learning. Since a pre-trained ResNet-50 was used, which was trained on a massive ImageNet database, this helped the network gain a feature extraction capability, thus making it possible to use it for medicinal leaf classification.

Hierarchical nature of feature learning is another technique that has been successfully implemented here. The lower layers, which were originally created to detect visual primitives such as edges, curves, and gradients in animals or vehicles, were successfully used to detect the same features in medicinal leaves (i.e., their margins, venations, and outlines). Higher layers have been used for generalization of information concerning the object represented by a leaf structure. Without transfer learning techniques, it is unlikely that a reasonable convergence would be reached in this case due to a limited amount of data used.

The "Black Box" & Texture vs. Shape:

However, despite its high efficiency, there are some specific characteristics of the algorithm, which are typical for deep learning algorithms, and that one is an inability to completely decipher how decisions are made. After investigating the reasons for mistakes we found that the algorithm makes use of texture-based features and general outline rather than any specific information about plant anatomy.

Botanists who work according to traditional principles recognize the type of plant based on various features such as presence/absence of stipules, attachment of petiole, hairiness (presence of trichomes), and veins' thickness. Nevertheless, due to the low resolution of the input image, 224 × 224 pixels, none of these features are available, thus making the algorithm rely solely on "Gestalt" principle of object recognition, meaning it recognizes the general outlines of leaves. Thus, it becomes understandable why morphologically alike plants like Jasmine and Karanda, which share the same sizes and general appearance, cannot be accurately recognized.

As we can see, there is a big difference between machine vision and human expertise in plant recognition. The former recognizes dominant characteristics while the latter is capable of utilizing

multisensory perception.

Environmental Robustness:

Another equally important component of this process involves the sensitivity of the algorithm to varying environmental factors. The most successful operation occurs in an instance where the leaf is easily identifiable and focused on a plain background. However, this may not be the case in reality. As a result, there might be a problem with the efficient function of this model in certain situations. There is always some level of impact of the following variables on the performance of this particular model. An example of the first variable involves the capture of the leaf in a camouflaged setting. For instance, the difficulty faced by the algorithms of edge detection and feature extraction in detecting the leaf against its background in the scenario of photographing a green Neem leaf surrounded by green bushes. The second variable includes any change in illumination. Particularly, a problem occurs with a waxed leaf being photographed using a flash that causes a specular reflection that hides the venation structure.

Scalability of the Knowledge Base:

Considering system design and engineering, the current solution employs static CSV storage for medicinal knowledge, which is easy to use but has drawbacks when considering scalability and maintenance.

At present, our system strongly depends on its dataset since adding another plant species to the database requires retraining the classifier, updating the CSV file, and changing the back-end code. In larger systems, it could be inconvenient because of possible human errors when performing these actions. Moreover, a static file is hardly convenient for implementing dynamic data queries, modifications, and processing.

Thus, a production-ready system should use the dynamic database approach, such as SQL or NoSQL databases. They could facilitate better data management, provide more flexibility regarding database operations, and enable scalable solutions. Also, a production-ready solution should separate its knowledge and classification models to enable independent updates of both elements.

All in all, this system is powerful, easy to implement, and has significant applicational value. At the same time, several aspects deserve further exploration to improve its effectiveness. They are related to interpreting the results of prediction, ensuring reliable classification in challenging environments, and improving scalability.

CHAPTER - 5

Conclusion and Future Study

5.1 Conclusion

The implementation of the project titled "A Unified Framework for Medicinal Leaf Classification Using Deep Learning and Recommendation Using TF-IDF and Logistic Regression" stands as a comprehensive demonstration of how modern artificial intelligence techniques can be effectively leveraged to preserve, enhance, and democratize traditional knowledge systems. By integrating computer vision, natural language processing, and full-stack development, this project successfully bridges the long-standing gap between the rich heritage of traditional medicinal practices, particularly Ayurveda, and the rapidly evolving digital ecosystem. One of the core challenges addressed by this work is the phenomenon of plant blindness, where individuals fail to recognize or appreciate the importance of plant species in everyday life. This issue has been effectively mitigated through an intelligent and automated system capable of identifying medicinal plants and presenting their real-world applications in an accessible manner.

From a technical standpoint, the project provides strong evidence supporting the effectiveness of transfer learning in solving domain-specific problems where large-scale annotated datasets are limited. By utilizing the ResNet-50 architecture pretrained on the ImageNet dataset, the system benefits from previously learned low-level and high-level feature representations such as edges, textures, and structural patterns. This significantly reduces the need for extensive computational resources and data collection efforts while still achieving high performance. The model demonstrates a validation accuracy of approximately 93%, indicating its ability to generalize effectively across diverse inputs and accurately classify around 80 different medicinal plant species. The hierarchical feature extraction capability of the network enables it to capture both fine-grained details like vein structures and broader characteristics such as leaf shape and orientation. Furthermore, optimization techniques such as the Adam optimizer, along with regularization strategies including dropout and data augmentation, enhance the model's robustness against variations in lighting, rotation, background noise, and environmental conditions.

A significant advancement introduced in the final stage of the project is the incorporation of a recommendation system powered by natural language processing. While image-based classification allows users to identify plants visually, the NLP component offers an alternative and highly intuitive interaction method. By employing TF-IDF vectorization and logistic regression, the system can interpret user-provided symptom descriptions and map them to relevant medicinal leaves. This dual-input approach—image and text—greatly enhances usability and accessibility, ensuring that users with varying levels of knowledge can benefit from the system. Additionally, the inclusion of detailed outputs such as usage methods, dosage guidelines, and precautionary measures transforms the system from a simple predictive tool into a comprehensive medicinal advisory platform.

Another noteworthy strength of this project lies in its practical implementation and deployment. Unlike many research efforts that remain confined to theoretical validation, this work extends into a fully functional application. The backend services are developed using a microservices architecture and deployed via FastAPI on Hugging Face Spaces, ensuring scalability and efficient handling of requests. On the frontend, a mobile application built using React Native (Expo) provides a seamless and user-friendly interface, allowing users to capture images, input symptoms, and receive results within seconds. The system demonstrates minimal latency, typically responding within 1–2 seconds, thereby ensuring a smooth user experience even under constrained network conditions.

The architectural design of the system reflects principles of modularity, scalability, and maintainability. By decoupling the image classification module, NLP recommendation engine, and frontend interface, the system allows for independent updates and future enhancements without affecting the overall functionality. RESTful APIs facilitate efficient communication between components, while the lightweight frontend ensures accessibility across a wide range of devices. This modular design not only improves performance but also lays a strong foundation for future expansion.

From a societal perspective, the impact of this project is highly significant. It contributes to the preservation and dissemination of traditional medicinal knowledge, making it readily accessible to a broader audience. By empowering individuals with information about medicinal plants and their uses, the system promotes self-reliance in managing minor health conditions and reduces

dependency on conventional healthcare systems for basic remedies. Furthermore, it serves as an educational tool for students and researchers, fostering a deeper understanding of the relationship between plant morphology and medicinal properties. Importantly, it also plays a role in safeguarding indigenous knowledge systems, ensuring that valuable cultural heritage is not lost in the process of modernization.

On a broader scale, this project exemplifies the successful integration of multiple domains—deep learning, natural language processing, and mobile application development—into a unified and functional system. It highlights how advanced technologies can be adapted to address real-world problems while remaining user-centric and culturally relevant. The work demonstrates that sophisticated AI systems can be designed not only for technical excellence but also for societal benefit and practical usability.

Looking ahead, this research opens up numerous opportunities for further enhancement. Future improvements may include expanding the dataset to include more plant species, incorporating multilingual support for wider accessibility, integrating more advanced NLP models such as transformers for improved text understanding, and enhancing explainability features to provide users with clearer insights into model predictions. Additionally, incorporating real-time environmental data or personalized health recommendations could further elevate the system's capabilities.

In conclusion, this project represents a significant step toward the development of intelligent, scalable, and socially impactful systems that effectively combine technology with traditional knowledge. It not only addresses a relevant real-world problem but also sets a strong foundation for future innovations in the field of AI-driven healthcare and botanical informatics.

5.2 Future Study

Although the current system demonstrates a robust and efficient framework for medicinal plant classification and recommendation, there remains significant scope for further enhancement and expansion. Future developments can transform this system from a functional prototype into a highly scalable, intelligent, and globally impactful platform. The following areas outline potential improvements and research directions.

- Edge AI and Offline Inference

One of the major limitations of the current system is its dependence on cloud-based inference, which requires a stable internet connection. In real-world scenarios, especially in rural, forest, or mountainous regions, network connectivity may not always be available. To overcome this limitation, future work can focus on implementing Edge AI, where the trained model is deployed directly on mobile devices.

This can be achieved by converting the trained PyTorch model into lightweight formats such as TensorFlow Lite or ONNX Runtime. Additionally, optimization techniques such as quantization (reducing 32-bit floating-point values to 8-bit integers) and pruning (removing redundant network parameters) can significantly reduce the model size and computational requirements. This would enable faster inference, lower power consumption, and complete offline functionality, thereby making the system more accessible and practical in remote environments.

- Real-Time Detection and Augmented Reality Integration

Currently, the system relies on static image inputs, requiring users to manually capture and upload leaf images. Future improvements can introduce real-time plant detection using advanced object detection models such as YOLO (You Only Look Once) or SSD (Single Shot Detector).

By integrating augmented reality (AR), the application can provide an interactive experience where users simply point their camera at plants, and the system overlays real-time information such as plant name, medicinal uses, and precautions directly onto the screen. This not only enhances usability but also transforms the application into an educational and exploratory tool for students, researchers, and general users.

- Multilingual and Vernacular Localization

To maximize the societal impact of the system, especially in rural areas where traditional knowledge is most prevalent, it is essential to eliminate language barriers. Future versions of the application can incorporate multilingual support, allowing users to access information in regional languages such as Telugu, Hindi, Tamil, Kannada, and Malayalam.

This enhancement would involve translating all system outputs, including disease descriptions, usage instructions, dosage information, and precautionary guidelines. Furthermore, integrating speech-based interfaces such as speech-to-text and text-to-speech can improve accessibility for

users with limited literacy, making the system inclusive and user-friendly.

- Geo-Tagging and Biodiversity Mapping

Future enhancements can include geo-tagging capabilities to collect location-based data of identified plants. With user consent, GPS data can be used to record where specific medicinal plants are found. Over time, this data can be aggregated to create biodiversity maps that visualize plant distribution across different regions.

This feature has significant implications for environmental monitoring, conservation efforts, and ecological research. It can help identify endangered species, track plant population trends, and support policymakers and forest departments in making informed decisions. Additionally, it promotes a crowdsourced approach to ecological intelligence, where users contribute to a larger knowledge network.

- Advanced Recommendation Systems and Personalization

The current recommendation system uses TF-IDF and logistic regression for symptom-based predictions. While effective, future improvements can incorporate more advanced machine learning and deep learning models to enhance recommendation accuracy and personalization.

By analyzing user interaction history, preferences, and feedback, the system can evolve into a personalized recommendation engine. Context-aware suggestions, adaptive learning mechanisms, and user profiling can improve the relevance of recommendations. This will enhance user engagement and provide a more tailored and meaningful experience.

- Integration with Healthcare Ecosystems

Another promising direction is the integration of the system with digital healthcare platforms. The application can be extended to support features such as telemedicine consultations, Electronic Health Record (EHR) integration, and collaboration with licensed healthcare professionals.

This would enable the system to function as a supportive clinical decision-making tool, providing preliminary insights while maintaining ethical boundaries. Such integration can significantly increase the credibility and practical utility of the system in real-world healthcare scenarios.

- Explainable AI (XAI) for Transparency

Deep learning models often operate as black boxes, making it difficult for users to understand how predictions are made. To address this issue, future work can incorporate Explainable AI (XAI) techniques such as Grad-CAM or attention visualization.

These methods can highlight specific regions of the leaf image that influenced the model's decision, providing visual explanations for predictions. This transparency will increase user trust, improve system interpretability, and make the model more acceptable in critical applications.

- Dataset Expansion and Knowledge Enrichment

The performance and robustness of the system can be further improved by expanding the dataset to include a larger variety of medicinal plants, environmental conditions, and seasonal variations. Incorporating images from diverse sources will enhance the model's generalization capability.

Additionally, collaboration with botanical experts, Ayurvedic practitioners, and domain specialists can ensure the accuracy and authenticity of medicinal information. This will strengthen the reliability of the system and make it more valuable for both educational and practical use.

- Scalability and System Optimization

As the number of users increases, it becomes essential to ensure that the system can handle large-scale interactions efficiently. Future improvements can focus on optimizing backend performance, reducing response time, and improving API efficiency.

Techniques such as load balancing, distributed computing, and caching mechanisms can be implemented to enhance scalability. This will ensure that the system remains responsive and reliable even under heavy usage conditions.

- Security, Privacy, and Ethical Considerations

With the collection of user data, including images, symptoms, and location information, it is important to address security and privacy concerns. Future versions of the system can incorporate encryption, secure authentication mechanisms, and data anonymization techniques.

Additionally, ethical considerations such as responsible AI usage, avoidance of misinformation,

and clear disclaimers regarding medical advice should be implemented to ensure safe and trustworthy usage of the system.

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APPENDIX

1. **Model Loading & Initialization Pipeline**

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = models.resnet50()
num_ftrs = model.fc.in_features
model.fc = nn.Sequential(
    nn.Linear(num_ftrs, 1024),
    nn.ReLU(),
    nn.Dropout(0.5),
    nn.Linear(1024, num_classes)
)
model.load_state_dict(torch.load(model_path, map_location=device))
model = model.to(device)
model.eval()
```

This block initializes the ResNet-50 architecture, modifies the final fully connected layer for custom classification, loads pretrained weights, and prepares the model for inference. It forms the core image classification pipeline.

2. **Image Preprocessing Pipeline**

```
val_transforms = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor(),
    transforms.Normalize(
        [0.485, 0.456, 0.406],
        [0.229, 0.224, 0.225]
    )
])
```

This defines the preprocessing pipeline applied to input images before feeding them into the model. It ensures consistency with training conditions.

3. **Core Inference Pipeline (Image → Prediction → Data)**

```
def classify_and_get_info(image_bytes):
    image = Image.open(io.BytesIO(image_bytes))

    if image.mode != 'RGB':
        image = image.convert('RGB')

    img_t = val_transforms(image)
    batch_t = torch.unsqueeze(img_t, 0).to(device)

    with torch.no_grad():
        out = model(batch_t)
        probabilities = torch.nn.functional.softmax(out, dim=1)[0]

    top_prob, top_idx = torch.max(probabilities, 0)

    predicted_class = class_names[top_idx.item()]
```

This function represents the end-to-end inference pipeline, where:

- Image is received
- Preprocessed
- Passed through the model
- Prediction + confidence extracted

4. Linking Model Output to Dataset (Knowledge Pipeline)

```
info = properties_df[properties_df['leaf_name'] == predicted_class]

result = {
    "prediction": predicted_class,
    "confidence": f"{top_prob.item():.2%}"
}

if not info.empty:
```

```
result["scientific_name"] = info.iloc[0]['scientific_name']
result["medicinal_uses"] = info.iloc[0]['medicinal_uses']
```

This connects the AI prediction → real-world medicinal knowledge, forming the intelligent recommendation pipeline.

5. NLP Recommendation Pipeline

```
user_tfidf = vectorizer.transform([user_input])
probabilities = text_model.predict_proba(user_tfidf)[0]

top_indices = np.argsort(probabilities)[-top_k:][::-1]
```

This snippet converts user symptoms into TF-IDF vectors and predicts relevant medicinal leaves using Logistic Regression.

6. Recommendation Data Mapping

```
leaf_info_match = disease_df[disease_df['Leaf'] == leaf_name]

usage = leaf_info.get('Usage', 'N/A')
dosage = leaf_info.get('Dosage', 'N/A')
precautions = leaf_info.get('Precautions', 'N/A')
```

Maps predicted leaf → usage, dosage, precautions → making system practically useful

7. Flask API Endpoint: Root

```
@app.route('/')
def index():
    return "Medicinal Leaf Classifier Server is running!"
```

Basic endpoint to verify backend is active.

8. Flask API Endpoint: Image Classification

```
@app.route('/predict', methods=['POST'])
def predict():
file = request.files['file']
img_bytes = file.read()
```

```
    result = classify_and_get_info(img_bytes)
    return jsonify(result)
```

This endpoint exposes the image classification pipeline as a REST API, enabling frontend integration.

9. Flask API Endpoint: NLP Recommendation

```
@app.route('/recommend', methods=['POST'])
def recommend():
    data = request.get_json(force=True)
    user_input = data.get('user_input')

    user_tfidf = vectorizer.transform([user_input])
    probabilities = text_model.predict_proba(user_tfidf)[0]
```

This endpoint enables symptom-based recommendation system, completing the dual-input architecture.

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

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Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

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12% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- **65** Not Cited or Quoted 9%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **23** Missing Citation 3%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

10% Internet sources
7% Publications
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